TEB2043: DATA SCIENCE

JANUARY 2026

GROUP 12 PROJECT REPORT

PROJECT TITLE: ANALYZING LIFESTYLE AND DIETARY FACTORS INFLUENCING OBESITY LEVELS

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1. Executive Summary

Obesity is one of the most pressing and rapidly growing public health challenges of the 21st century, with its prevalence rising steadily across both developed and developing nations. Driven by increasingly sedentary lifestyles, high-calorie dietary habits and growing technology dependence. Addressing this challenge requires moving beyond generalized public health messaging and toward a more precise, data-driven understanding of exactly which daily behaviors place individuals at greatest risk.

This project "*Analyzing Lifestyle and Dietary Factors Influencing Obesity Levels"* was designed to do precisely that. Using a structured dataset of 2,111 individuals sourced from Kaggle, comprising 17 attributes that capture dietary patterns, physical activity, transportation habits and demographic characteristics, the study applied a full data science methodology to build and compare five machine learning classification models which are Logistic Regression, Decision Tree, K-Nearest Neighbors (KNN), Support Vector Machine (SVM) and Random Forest. The core analytical objective was to determine whether an individual's obesity level which is classified across seven distinct categories ranging from Insufficient Weight to Obesity Type III could be reliably predicted from their lifestyle habits alone, deliberately excluding direct physical measurements such as height and weight to prevent target leakage and ensure the models learned from behavior rather than biometrics.

The data preparation pipeline was thorough and methodologically sound. Twenty-four duplicate records were removed, extreme outliers in key features were controlled through Winsorization, all categorical variables were numerically encoded, and a rigorous ANOVA F-test was applied to eliminate statistically non-significant features. The result was a clean, 14-feature training dataset that balanced predictive power with model efficiency.

Across all five models evaluated, the Random Forest classifier emerged as the strongest performer with a test accuracy of 87.95%. This demonstrates that the collective judgment of 100 independent decision trees is the most effective strategy for modeling the complex and non-linear relationships between lifestyle behaviors and obesity outcomes. Feature importance analysis consistently identified age, vegetable consumption frequency, screen time and physical activity as the most influential predictors of weight classification.

A key finding shared across all models was the universal challenge of distinguishing between individuals in the middle weight categories which are Normal Weight, Overweight Level I and Overweight Level II. This reflects the genuine behavioral similarity between these groups and highlights the importance of capturing finer-grained lifestyle data in future studies. Conversely, the most severe obesity categories were reliably identified by every model tested, confirming that extreme weight classifications carry distinctly recognizable behavioral signatures

The results of this study carry both technical and human significance. They demonstrate that machine learning models trained on everyday lifestyle data can meaningfully predict obesity risk for practical applications in preventive healthcare, early clinical screenin and the design of more targeted public health interventions. With further development, refinement, and deployment, tools built on this foundation have the potential to reach individuals before risk becomes reality.

2. Project Description

2.1. Problem Description

The increasing prevalence of obesity worldwide is an urgent public health challenge, significantly driven by sedentary lifestyles, high-calorie diets and growing technology dependence. Unhealthy weight gain is associated with a heightened risk of severe chronic diseases, including diabetes, hypertension, and cardiovascular conditions. To effectively combat this issue, it is critical to move beyond generalized health advice and systematically analyze the specific behaviors and routines that impact body weight.

2.2. Business Goal

The primary business objective of this project is to support early obesity risk detection and promote preventive healthcare strategies by understanding exactly how daily habits influence long-term health outcomes. By uncovering the major lifestyle and dietary contributors to obesity, this analysis aims to provide practical insights that inform healthier lifestyle recommendations. Ultimately, these data-driven insights are designed to assist policymakers in designing targeted health awareness programs and to empower individuals to understand their personal lifestyle risk factors.

2.3. Technical Goal

To achieve the business objectives, the technical goal of this project centers on multiclass classification predictive modeling. The project will process and analyze a structured Obesity Dataset obtained from Kaggle to investigate the statistical relationships between lifestyle features and body weight. The technical workflow will implement full data science lifecycle methodologies which includes exploratory data analysis, data preparation and feature selection to identify the most influential variables affecting obesity classification.

Specifically, the project will build, evaluate, and compare five machine learning classification algorithms which are Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN) and Support Vector Machines (SVM). These models will be trained to categorize individuals into one of seven distinct obesity levels, ranging from Insufficient Weight to Obesity Type III. Model performance will be assessed using standard evaluation metrics (Accuracy, Precision, Recall, F1-score, and Confusion Matrix) to determine the most highly accurate predictive model. Finally, the technical deliverable includes developing an interactive dashboard to visually present feature importance, model performance and key dietary trends to technical and non-technical audiences.

3. Data Description

3.1. Source of Dataset

The dataset utilized for this analysis is the "Obesity Prediction Dataset" obtained from the Kaggle open-data platform, authored by Adeniran Stephen. It is a structured, cross-sectional dataset comprising 2,111 original observations and 17 attributes, covering individuals from Mexico, Peru and Colombia. The dataset captures lifestyle and health-related information including eating habits, family history and physical condition, with labels classifying obesity levels ranging from underweight to different obesity types. This dataset was chosen because:

- It focuses directly on lifestyle and dietary behaviours
- It allows classification and predictive modeling.
- It aligns with real-world health analytics applications.

3.2. Schematic View of the Dataset

The dataset consists of a combination of continuous numerical and categorical variables detailing dietary habits, physical activity, and demographic information. Below is the data dictionary defining the 17 core attributes:

Variable	Description
Gender	Gender of Individual
Age	Age in years
Height	Height (meters)
Weight	Weight (kg)
FAVC	Frequent consumption of high-calorie food (Yes/No)
FCVC	Frequency of vegetable consumption
NCP	Number of main meals per day
CAEC	Eating between meals
CH2O	Daily water consumption
FAF	Physical activity frequency
TUE	Time spent using technology devices
CALC	Alcohol consumption frequency
SMOKE	Smoking habits
MTRANS	Mode of transportation
NObeyesdad	Obesity level classification **(Target Variable)**

*(**Table 1.1:** Details the specific features included in the dataset.)*

The Target Variable: The primary focus of this predictive analysis is the **NObeyesdad** attribute, which serves as our target variable. This categorical variable represents an individual's classification of body weight based on their lifestyle and physical metrics. It is divided into seven distinct risk categories:

1. Insufficient Weight
2. Normal Weight
3. Overweight Level I
4. Overweight Level II
5. Obesity Type I
6. Obesity Type II
7. Obesity Type III

Description of risk categories:

1. **Insufficient Weight (BMI < 18.5)**
    - Body weight is below the healthy range and may increase health risks.
2. **Normal Weight (BMI 18.5–24.9)**
    - Body weight is within the healthy range with lower risk of disease.
3. **Overweight Level I (BMI 25.0–27.4)**
    - Body weight is slightly above normal and may increase risk of health problems.
4. **Overweight Level II (BMI 27.5–29.9)**
    - Body weight is high and associated with greater risk of chronic diseases.
5. **Obesity Type I (BMI 30.0–34.9)**
    - Mild obesity with significant risk of cardiovascular and metabolic diseases.
6. **Obesity Type II (BMI 35.0–39.9)**
    - Moderate obesity with very high risk of serious health conditions.
7. **Obesity Type III (BMI ≥ 40)**
    - Severe obesity with extremely high health risk and possible need for medical treatment.

4. Data Understanding and Exploration (EDA)

4.1. Data Completeness and Duplicates

An initial assessment confirmed that the dataset was highly complete, with 0 missing values found across all 17 attributes. However, an analysis of the 2,111 raw records identified 24 exact duplicate rows. These duplicated observations pose a risk of causing machine learning models to overfit or memorize repeated information if left unaddressed.

```
> total_na <- sum(is.na(df))
> cat(sprintf("-> Missing Values Check: Found %d total missing values.\n", total_na))
-> Missing Values Check: Found 0 total missing values.
```

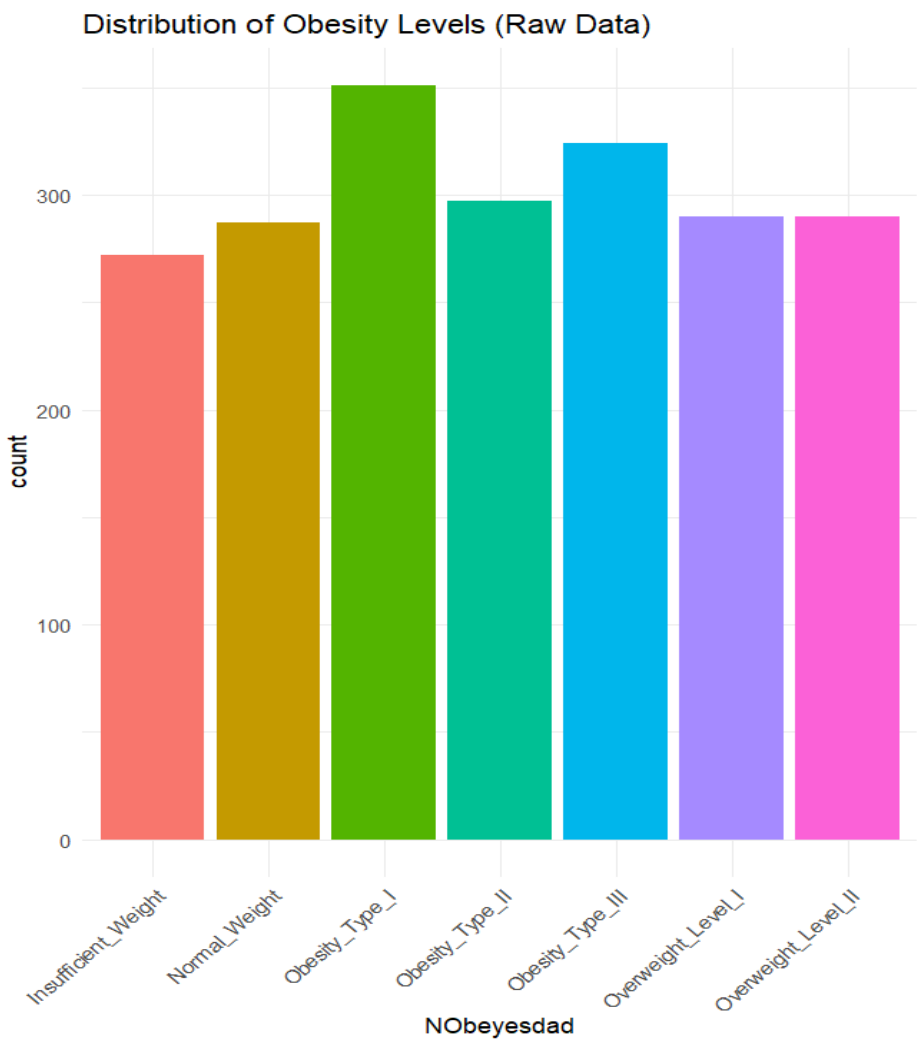
***(Figure 2.1:** Shows output of checking for missing values in the dataset)*

```
   [PROOF] Found exactly 24 duplicate rows. Here are ALL of them:
# A tibble: 24 × 5
   Gender   Age Height Weight NObeyesdad
   <chr>  <dbl>  <dbl>  <dbl> <chr>
 1 Female    21   1.52     42 Insufficient_Weight
 2 Female    25   1.57     55 Normal_Weight
 3 Male      21   1.62     70 Overweight_Level_I
 4 Male      21   1.62     70 Overweight_Level_I
 5 Male      21   1.62     70 Overweight_Level_I
 6 Female    22   1.69     65 Normal_Weight
 7 Female    16   1.66     58 Normal_Weight
 8 Female    18   1.62     55 Normal_Weight
 9 Male      22   1.74     75 Normal_Weight
10 Male      18   1.72     53 Insufficient_Weight
11 Female    21   1.52     42 Insufficient_Weight
12 Female    21   1.52     42 Insufficient_Weight
13 Female    21   1.52     42 Insufficient_Weight
14 Male      21   1.62     70 Overweight_Level_I
15 Male      21   1.62     70 Overweight_Level_I
16 Male      21   1.62     70 Overweight_Level_I
17 Male      21   1.62     70 Overweight_Level_I
18 Male      21   1.62     70 Overweight_Level_I
19 Male      21   1.62     70 Overweight_Level_I
20 Male      21   1.62     70 Overweight_Level_I
21 Male      21   1.62     70 Overweight_Level_I
22 Male      21   1.62     70 Overweight_Level_I
23 Male      21   1.62     70 Overweight_Level_I
24 Male      21   1.62     70 Overweight_Level_I
>
```

***(Figure 2.2:** Shows list of duplicate rows in the dataset)*

4.2. Target Variable Distribution

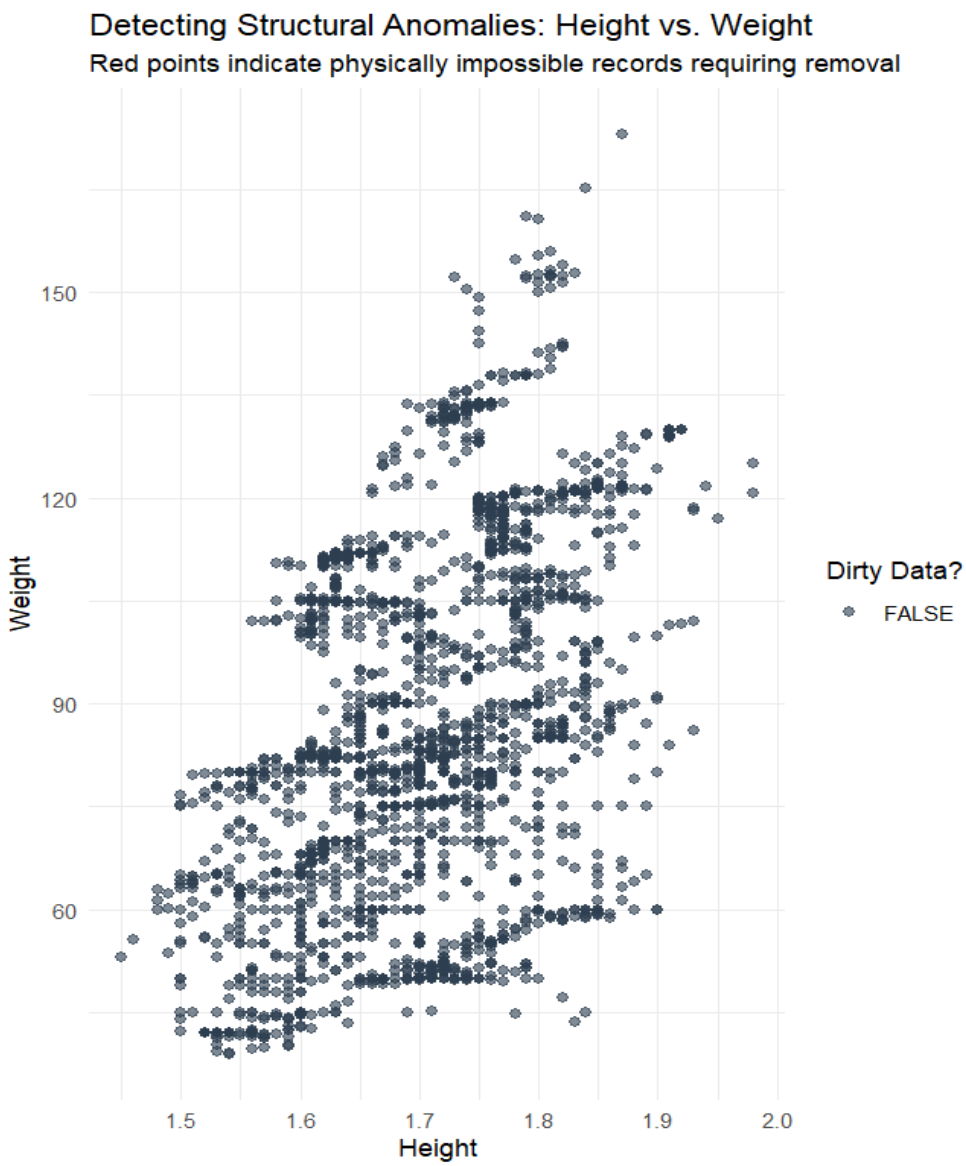
The target variable, Nobeyesdad (obesity level), was visualized to check even distribution of data across all seven categories. The calculated imbalance ratio was 1.30. Because this value falls well below the standard warning threshold of 1.5, the dataset's target variable was considered naturally balanced and healthy.



***(Figure 2.3:** Shows the distribution of obesity types in the dataset)*

4.3. Structural Anomaly Detection

The dataset was evaluated for physically impossible values, such as negative ages, negative heights, or a Body Mass Index (BMI) of 80 or above. **Because BMI was not included in the original raw dataset, a temporary BMI column was mathematically engineered (Weight / Height^2) specifically to act as a diagnostic tool.** A scatter plot comparing Height and Weight, alongside a BMI distribution histogram, was generated. The diagnostic checks confirmed there were 0 severe structural errors, meaning the baseline physical measurements are safe and reliable for analysis.



***(Figure 2.4:** Scatter plot of weight against height)*

4.4. Categorical Skewness (Transportation Habits)

An analysis of the transportation mode column (MTRANS) revealed a heavy imbalance. A significant majority of individuals used "Public_Transportation" (1,580) or an "Automobile" (457), while very few chose active methods like walking (56) or biking (7). This skewed distribution justified the decision to later engineer this variable into two simpler categories: active transportation and passive transportation.

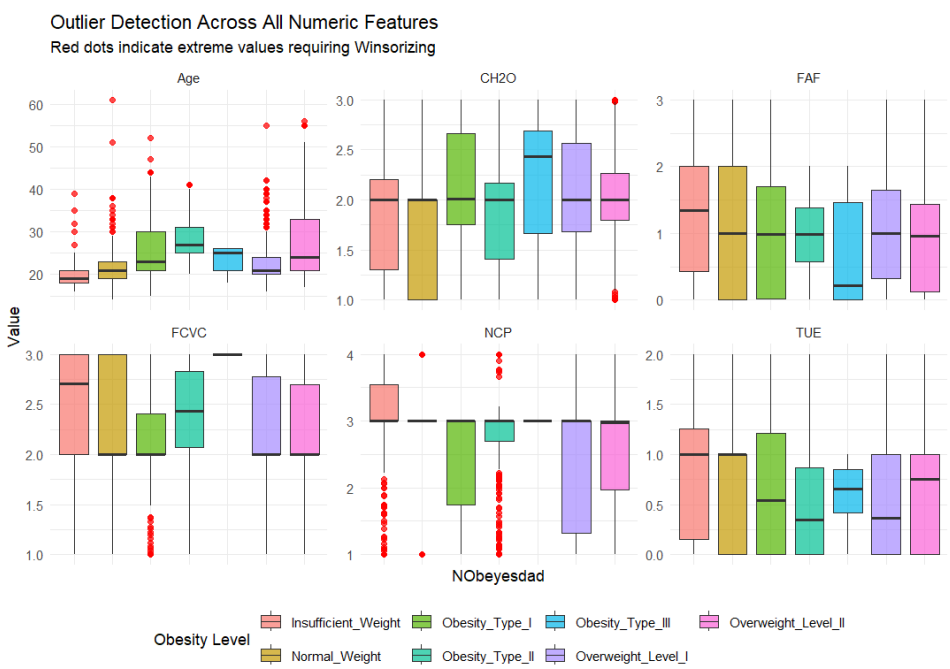
```
> # Audit B: Categorical Skewness (MTRANS)
> cat("-> Transportation (MTRANS) Distribution (Justifies Feature Engineering):\n")
-> Transportation (MTRANS) Distribution (Justifies Feature Engineering):
> print(table(df$MTRANS))

          Automobile                Bike           Motorbike Public_Transportation             Walking
                 457                   7                  11                 1580                  56
>
```

***(Figure 2.5:** Distribution of diffent classes in the MTRANS/Transportation Feature)*

4.5. Outlier Identification

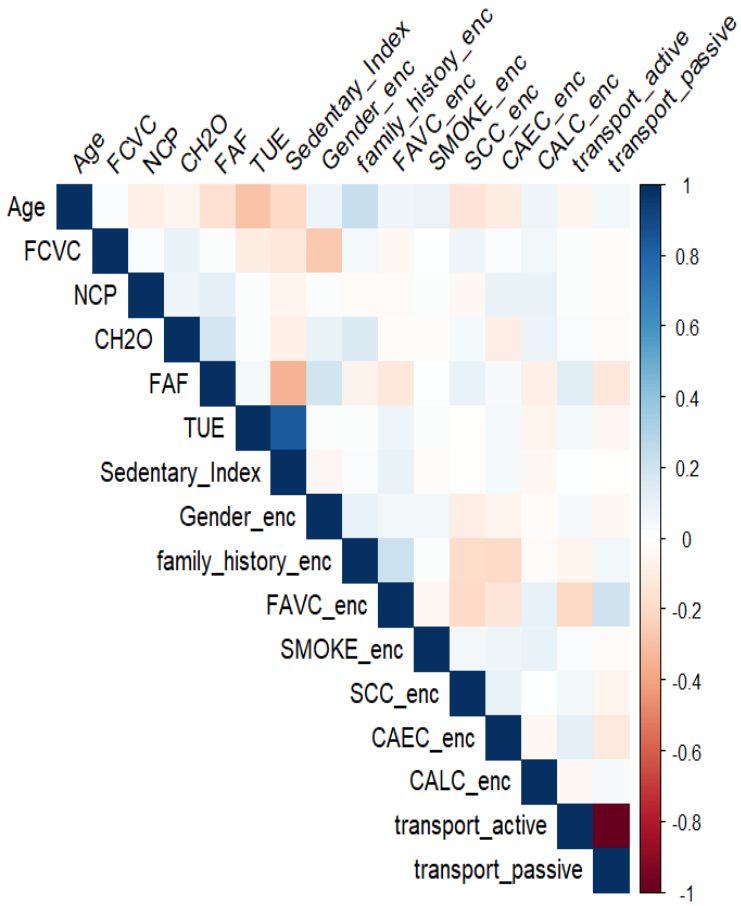
Boxplots were used to examine numerical features for extreme values. The visualizations displayed significant outliers that fell far outside the normal range. Specifically, the code identified exactly **128 extreme values in the Age feature** and **481 in the NCP (meals per day) feature**. This evidence supported the decision to use a technique called Winsorizing in the preparation phase. Winsorizing caps these extreme values, so they do not negatively distort the machine learning models.



***(Figure 2.5:** Boxplot of some features in the dataset)*

4.6. Feature Correlation and Statistical Significance

A correlation heatmap and an ANOVA F-test were utilized to evaluate how mathematically useful each raw feature is for predicting obesity. The analysis identified high collinearity (>0.85) between the engineered active and passive transportation features, meaning they provide redundant information. As a result, the redundant feature was flagged to be dropped to simplify and improve the efficiency of the models.



(***Figure 2.6:*** *Correlation heatmap of dataset features*)

4.7. Non-Numerical Feature Data

A review of the dataset's structure revealed that exactly eight independent features are stored as non-numerical text (strings). These consist of binary categorical variables (Gender, family_history_with_overweight, FAVC, SMOKE, SCC) and multi-class categorical variables (CAEC, CALC, MTRANS). Because machine learning algorithms rely strictly on mathematical computations and distance metrics (like in SVM or KNN), they cannot natively process raw text. These eight non-numerical features represent a critical structural incompatibility that must be resolved through numerical encoding prior to model training.

	Gender	family_history_with_overweight	FAVC	SMOKE	SCC	CAEC	CALC	MTRANS
	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>	<chr>
1	Female	yes	no	no	no	Sometimes	no	Public_Transportation
2	Female	yes	no	yes	yes	Sometimes	Sometimes	Public_Transportation
3	Male	yes	no	no	no	Sometimes	Frequently	Public_Transportation
4	Male	no	no	no	no	Sometimes	Frequently	Walking
5	Male	no	no	no	no	Sometimes	Sometimes	Public_Transportation
6	Male	no	yes	no	no	Sometimes	Sometimes	Automobile
7	Female	yes	yes	no	no	Sometimes	Sometimes	Motorbike
8	Male	no	no	no	no	Sometimes	Sometimes	Public_Transportation
9	Male	yes	yes	no	no	Sometimes	Frequently	Public_Transportation
10	Male	yes	yes	no	no	Sometimes	no	Public_Transportation

(***Figure 2.7:*** *Values of Non-Numerical Features*)

4.8. Feature Scaling/Standardization

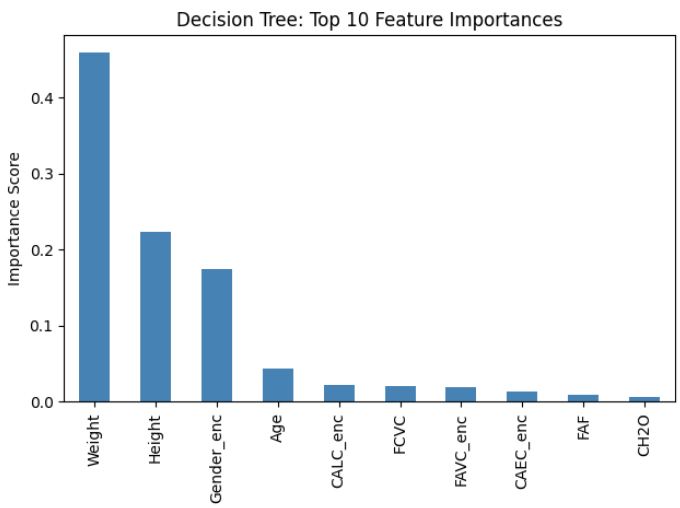
An examination of the eight continuous numerical features (Age, Height, Weight, FCVC, NCP, CH2O, FAF, and TUE) revealed significant differences in their baseline scales and ranges. For example, the physical variables (Age, Height, and Weight) span across tens or hundreds of units. Conversely, the survey-based lifestyle variables (FCVC, NCP, CH2O, FAF, and TUE) are strictly constrained to a much smaller 0-to-3 scale. Because distance-based and gradient-based machine learning algorithms (such as SVM, KNN, and Logistic Regression) calculate mathematical distances between data points, leaving these features unstandardized would cause the algorithms to incorrectly assign higher importance to the physical features simply because they have larger raw magnitudes. This scaling disparity across all eight numerical features must be mathematically addressed prior to model training.

	Age	Height	Weight	FCVC	NCP	CH2O	FAF	TUE
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	21	1.62	64	2	3	2	0	1
2	21	1.52	56	3	3	3	3	0
3	23	1.8	77	2	3	2	2	1
4	27	1.8	87	3	3	2	2	0
5	22	1.78	89.8	2	1	2	0	0
6	29	1.62	53	2	3	2	0	0
7	23	1.5	55	3	3	2	1	0
8	22	1.64	53	2	3	2	3	0
9	24	1.78	64	3	3	2	1	1
10	22	1.72	68	2	3	2	1	1

(***Figure 2.8:*** *Values of Numerical Features*)

4.9. Target Leakage

During the exploration phase, a preliminary feature importance test was conducted using a Decision Tree algorithm to observe how the dataset variables interacted. The resulting Feature Importances audit revealed a severe case of "Target Leakage." The algorithm assigned the vast majority of its predictive power, a combined 68.21%, directly to the underlying physical measurements: Weight (45.86%) and Height (22.35%). By exploiting these two physical variables, the model calculated its way to an artificially high 93.98% accuracy while almost entirely ignoring the actual dietary and behavioral factors this study aims to analyze (such as physical activity FAF at just 0.96% importance, or water intake CH2O at 0.64%). Because the target variable (Nobeyesdad) is inherently a classification of body mass, leaving Weight and Height in the dataset allows the model to "cheat" and predict the outcome without learning the underlying lifestyle causes. To ensure the final machine learning models are forced to evaluate the true behavioral causes of obesity, both Weight and Height must be strictly excluded from the final training features.




```
==================================================
         Top 10 Feature Importances
==================================================
  Weight                : 0.4586
  Height                : 0.2235
  Gender_enc            : 0.1737
  Age                   : 0.0433
  CALC_enc              : 0.0226
  FCVC                  : 0.0203
  FAVC_enc              : 0.0198
  CAEC_enc              : 0.0130
  FAF                   : 0.0096
  CH2O                  : 0.0064
==================================================
```

(***Figures 2.9:*** *Barchart and list of most significant features*)

4.10. Ordinal vs. Nominal Variable Hierarchies

While reviewing the non-numerical categorical variables, a critical structural distinction was identified. Some variables, such as Gender or FAVC (Frequent consumption of high-calorie food – yes/no), are purely *nominal* and have no inherent mathematical ranking. However, variables like CAEC (Consumption of food between meals) and CALC (Consumption of alcohol) contain text responses like "No," "Sometimes," "Frequently," and "Always." These represent *ordinal* data with a strict logical hierarchy. Recognizing this distinction during the EDA phase is crucial, as ordinal variables must be deliberately mapped to sequential integers (e.g., 0 to 3) during data preparation to preserve their hierarchical magnitude for the algorithms.

4.11. Feature Imbalance Analysis

An evaluation of the categorical features revealed a severe imbalance in three lifestyle habits: Smoking (SMOKE), Calorie Counting (SCC), and Passive Transportation. In each case, a single category makes up over 95% of the data. However, this extreme imbalance is not an issue because these are independent variables, not the target variable. Since data-balancing techniques are only required for the target variable (Nobeyesdad) to ensure the model learns fairly, these imbalanced independent features were left in their natural state.

```
Feature: transport_passive

    0     1
 2.63 97.37
⚠️ WARNING: Highly imbalanced! Dominant category is 97.37%
--------------------------------------------------
--------------------------------------------------
Feature: SMOKE_enc

    0     1
97.61  2.39
⚠️ WARNING: Highly imbalanced! Dominant category is 97.61%
--------------------------------------------------
Feature: SCC_enc

    0     1
95.22  4.78
⚠️ WARNING: Highly imbalanced! Dominant category is 95.22%
--------------------------------------------------
```

(***Figure 2.10:****List of features with unbalanced observation*)

5. Data Preparation

5.1. Data Cleaning

The first step was to remove identical rows and structural errors that could confuse the models. The code identified and removed 24 duplicate rows. It also filtered the data to ensure all records were physically possible (Age > 0, Height > 0, Weight > 0, and a calculated Body Mass Index below 80).

```
> n_dupes <- nrow(df) - nrow(distinct(df))
> df <- distinct(df)
>
> df$BMI <- df$Weight / (df$Height ^ 2)
> before_rows <- nrow(df)
>
> df <- df %>%
+   filter(Age > 0 & Height > 0 & Weight > 0 & BMI < 80)
>
> cat(sprintf("[CLEAN] Step 6: Removed %d duplicates & %d structural errors.\n", n_dupes, before_rows - nrow(df)))
[CLEAN] Step 6: Removed 24 duplicates & 0 structural errors.
```

(***Figure 3.1:*** *Result of removing duplicate rows and ensure all rows contain physically possible values*)

5.2. Feature Engineering and Encoding

Machine learning algorithms require numerical input, so all categorical (text-based) data was converted into numbers:

- **Target Encoding:** The main Nobeyesdad target variable was encoded from 0 to 6 to represent the seven obesity levels.

```
# 1. Target Encoding
NObeyesdad_enc = as.integer(factor(NObeyesdad, levels = obesity_levels)) - 1,
```

	NObeyesdad	NObeyesdad_enc
	<chr>	<dbl>
1	Normal_Weight	1
2	Normal_Weight	1
3	Normal_Weight	1
4	Overweight_Level_I	2

(***Figure 3.2:*** *Result of encoded Nobeyesdad target variable*)

- **Binary Encoding:** The five binary variables (Gender, family_history_with_overweight, FAVC, SMOKE, and SCC) were mathematically converted into 0 and 1.

```r
# 2. Binary Encoding
Gender_enc = as.integer(as.factor(Gender)) - 1,
family_history_enc = as.integer(as.factor(family_history_with_overweight)) - 1,
FAVC_enc = as.integer(as.factor(FAVC)) - 1,
SMOKE_enc = as.integer(as.factor(SMOKE)) - 1,
SCC_enc = as.integer(as.factor(SCC)) - 1,
```

	Gender_enc	family_history_enc	FAVC_enc	SMOKE_enc	SCC_enc
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	0	1	0	0	0
2	0	1	0	1	1
3	1	1	0	0	0
4	1	0	0	0	0

(***Figure 3.2:*** *Result of binary encoding features that were originally non-numeric*)

- **Ordinal Encoding:** The frequency-based survey variables (CAEC and CALC) were deliberately mapped to a sequential ordinal scale (0 = No, 1 = Sometimes, 2 = Frequently, 3 = Always) to preserve their logical hierarchy.

```r
# 3. Ordinal Encoding
CAEC_enc = ordinal_map[CAEC],
CALC_enc = ordinal_map[CALC],
```

	CAEC_enc	CALC_enc
	<dbl>	<dbl>
1	1	0
2	1	1

(***Figure 3.3:*** *Result of ordinal encoding features that were originally non-numeric*)

- **Feature Engineering:** Because the EDA phase revealed extreme skewness in transportation habits, the MTRANS column was transformed into two simpler binary columns: transport_active (walking/biking) and transport_passive (automobiles/public transit).

```
# 4. Feature Engineering
transport_active = ifelse(MTRANS %in% c('Walking','Bike'),1,0),
transport_passive = ifelse(MTRANS %in% c('Automobile','Motorbike','Public_Transportation'),1,0)
```

transport_active	transport_passive
<dbl>	<dbl>
0	1
0	1
0	1
1	0

(***Figure 3.4:*** *Result of feature engineering the MTRANS feature*)

5.3. Data Splitting (Preventing Data Leakage)

Before applying any statistical transformations, the dataset was split into two separate groups: a Training set (80%) and a Testing set (20%). Splitting the data first acts as a firewall, ensuring that the model does not accidentally "peek" at the testing data while it is learning.

```
> set.seed(42)
> trainIndex <- createDataPartition(y, p = 0.8, list = FALSE)
>
> X_train <- X[trainIndex,]
> X_test  <- X[-trainIndex,]
> y_train <- y[trainIndex]
> y_test  <- y[-trainIndex]
>
> cat(sprintf("[SPLIT] Step 9: Train rows: %d | Test rows: %d\n", nrow(X_train), nrow(X_test)))
[SPLIT] Step 9: Train rows: 1672 | Test rows: 415
>
```

(***Figure 3.5:*** *Result of splitting the dataset*)

5.4. Handling Outliers and Imputation

To address the extreme values found in the EDA, **Winsorizing** was applied specifically to **Age** (128 outliers) and **NCP** (481 outliers). These values were capped at the 1.5x IQR boundary. While the initial audit found no missing data, a median default value was programmed into the pipeline to guarantee that any future missing values would be safely replaced the median value.

```
> for (col in num_cols) {
+   Q1 <- quantile(X_train[[col]], 0.25, na.rm=TRUE)
+   Q3 <- quantile(X_train[[col]], 0.75, na.rm=TRUE)
+   IQR <- Q3 - Q1
+   lower <- Q1 - 1.5 * IQR
+   upper <- Q3 + 1.5 * IQR
+
+   # Diagnostic check: count how many are out of bounds in train set
+   outliers_capped <- sum(X_train[[col]] < lower | X_train[[col]] > upper, na.rm=TRUE)
+   cat(sprintf("   -> %s: %d outliers capped\n", col, outliers_capped))
+
+   X_train[[col]] <- pmin(pmax(X_train[[col]], lower), upper)
+   X_test[[col]]  <- pmin(pmax(X_test[[col]], lower), upper)
+ }
   -> Age: 128 outliers capped
   -> FCVC: 0 outliers capped
   -> NCP: 481 outliers capped
   -> CH2O: 0 outliers capped
   -> FAF: 0 outliers capped
   -> TUE: 0 outliers capped
>
```

(***Figure 3.6:*** *Result of removing outliers in some features of the dataset*)

5.5. Feature Scaling/Z-score Normalization

To resolve the scaling disparity between large physical measurements and smaller survey-based scales, **Standardization** (also known as **Z-score Normalization**) was applied. Using the StandardScaler method, all six continuous features—Age, FCVC, NCP, CH2O, FAF, and TUE—were mathematically transformed to have a mean of 0 and a standard deviation of 1. This process ensures that features with larger raw magnitudes, such as Weight or Age, do not disproportionately influence the distance-based calculations of models like SVM and KNN. By placing all numerical variables on a uniform scale, the machine learning algorithms are forced to evaluate the predictive importance of each lifestyle and dietary factor fairly.

```
scale_cols <- c('Age','FCVC','NCP','CH2O','FAF','TUE')
scaler <- preProcess(X_train[,scale_cols], method=c("center","scale"))
```

	Age	FCVC	NCP	CH2O	FAF	TUE
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	-0.565	-0.778	0.393	0.0122	-1.19	0.559
2	-0.565	1.08	0.393	1.66	2.30	-1.09
3	-0.190	-0.778	0.393	0.0122	1.14	0.559
4	0.560	1.08	0.393	0.0122	1.14	-1.09
5	-0.378	-0.778	-1.70	0.0122	-1.19	-1.09
6	-0.190	1.08	0.393	0.0122	-0.0235	-1.09

(***Figure 3.7:*** *Result of standardizing the unstandardized features of the dataset*)

5.6. Feature Selection

To optimize the models, unnecessary or harmful features were removed using mathematical and logical tests:

- **Correlation Check:** The pipeline identified a perfect negative correlation between the engineered transport_active and transport_passive variables. To resolve this redundancy, transport_active was removed.
- **ANOVA F-Test:** A statistical test confirmed that all remaining behavioral and dietary features are significant predictors of obesity ().
- **Preventing Target Leakage (Dropping Height and Weight):** Based on the results of the EDA, **Height and Weight were purposefully excluded** from the final feature set. Because the target variable (Nobeyesdad) is a classification derived from Body Mass Index (BMI), keeping Height and Weight would allow the model to "cheat" by simply acting as a calculator. Removing these physical measurements forces the machine learning algorithms to identify the true lifestyle-based causes of obesity rather than just measuring body size.

6. Proposed Data Science Methodology

6.1. Data Analysis and Modeling Technique Used

To achieve the technical goal of predicting obesity risk based on lifestyle and dietary habits, a supervised machine learning approach was adopted. Because the target variable (Nobeyesdad) consists of seven distinct weight categories, the core analytical task was framed as a multiclass classification problem.

To determine the most accurate predictive algorithm for this specific dataset, five distinct classification models were developed, trained, and compared:

1. **Logistic Regression:** Selected as a robust statistical baseline model. It estimates the probability of an individual belonging to a specific obesity class using a linear combination of their lifestyle features. It was configured with the 'lbfgs' solver to efficiently handle the multiclass nature of the dataset.
2. **Decision Tree:** A non-linear model that categorizes data by creating a sequence of logical splits based on feature thresholds (e.g., separating individuals by high or low vegetable consumption). It was chosen for its high interpretability, which helps in understanding the underlying decision logic.
3. **Random Forest:** A powerful ensemble method that builds multiple independent decision trees and merges their predictions to output a final classification. It was chosen for its ability to capture complex, non-linear relationships in dietary behaviors while significantly reducing the risk of model overfitting.
4. **K-Nearest Neighbors (KNN):** A distance-based algorithm configured to evaluate the 5 closest data points (). It classifies a person's obesity level based on the majority class of the individuals who have the most mathematically similar lifestyle habits.
5. **Support Vector Machine (SVM):** Configured with a Radial Basis Function (RBF) kernel, this model was selected for its strong mathematical ability to draw complex, multidimensional boundaries between closely related obesity categories.

To ensure a fair and rigorous comparison, all five models were trained on the exact same optimized training dataset containing the 14 finalized lifestyle features. Their predictive performance was then evaluated on the unseen testing data using standard industry metrics: Accuracy, Precision, Recall, and the F1-Score. Additionally, a Confusion Matrix was utilized to visualize specific classification errors.

6.2. Model Implementation & Evaluation

6.2.1. Decision Tree

Code Implementation

The model was built using the DecisionTreeClassifier from the scikit-learn library. The following code demonstrates the training and evaluation process:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix


# Class label names mapped to target variable
LABEL_MAP = {
    0: "Insufficient Weight", 1: "Normal Weight", 2: "Overweight L1",
    3: "Overweight L2", 4: "Obesity T1", 5: "Obesity T2", 6: "Obesity T3",
}
CLASS_NAMES = [LABEL_MAP[i] for i in sorted(LABEL_MAP)]

# 1. Load and Prepare Data
train = pd.read_csv("/content/drive/MyDrive/Data Science Project/train.csv")
test  = pd.read_csv("/content/drive/MyDrive/Data Science Project/test.csv")

# Standardize boolean columns to integers
bool_cols = train.select_dtypes(include="bool").columns
train[bool_cols] = train[bool_cols].astype(int)
test[bool_cols]  = test[bool_cols].astype(int)

X_train = train.drop("NObeyesdad_enc", axis=1)
y_train = train["NObeyesdad_enc"]
X_test  = test.drop("NObeyesdad_enc", axis=1)
y_test  = test["NObeyesdad_enc"]

# 2. Train Model (Applying pre-pruning via max_depth)
dt = DecisionTreeClassifier(max_depth=8, random_state=42)
dt.fit(X_train, y_train)

# 3. Evaluate Performance
y_pred   = dt.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
```

(***Figure 3.1:*** *Implementation of Decision Tree Model*)

How the Model Works and Key Settings

A Decision Tree works a lot like a flowchart. It looks at the lifestyle data and asks a series of True/False questions (for example, "Does this person eat vegetables frequently?") to divide the data into the final weight categories. To make sure the model worked correctly for this project, the following settings were used:

- **max_depth=8 (Maximum Depth):**
  If a decision tree is allowed to ask too many questions, it will just memorize the exact individuals in the training data instead of learning general health patterns (a problem known as "overfitting"). Setting a limit of 8 levels forces the tree to stop growing early so it can still make accurate predictions on new, unseen people.
- **random_state=42:**
  This setting acts as a fixed starting point. It ensures that every time the code is run, the model splits the data the exact same way, making the results consistent and reproducible for this report.

Evaluation Strategy

The model's performance was evaluated by testing it against the unseen holdout dataset (415 samples). The predictions were then compared to the actual target values to generate the final accuracy, precision, recall, and F1-scores.

- **Overall Test Accuracy: 69.64%**

```
==================================================
         Decision Tree — Results
==================================================
  Overall Accuracy : 69.64%
  Training samples : 1672
  Test samples     : 415
==================================================

Classification Report:
                     precision    recall  f1-score   support

Insufficient Weight       0.81      0.70      0.75        54
      Normal Weight       0.59      0.58      0.59        55
      Overweight L1       0.67      0.45      0.54        58
      Overweight L2       0.45      0.64      0.53        55
         Obesity T1       0.66      0.73      0.70        75
         Obesity T2       0.77      0.74      0.75        54
         Obesity T3       1.00      0.98      0.99        64

           accuracy                           0.70       415
          macro avg       0.71      0.69      0.69       415
       weighted avg       0.71      0.70      0.70       415

Test Accuracy: 69.64 %
CV Accuracy: 68.13 %
```

(***Figure 3.2:*** *Evaluation of Decision Tree Model*)

6.2.2. Random Forest

Code Implementation

The model was built using the RandomForestClassifier class from the scikit-learn library. The following code demonstrates the training and evaluation process:

```python
import pandas as pd
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report

# ── Load Data ──────────────────────────────────────────────────────────
train = pd.read_excel("C:/Users/User/Desktop/UTP/UG/2nd Year/2nd Semester/Data Sceince/Project/train.csv.xlsx")
test  = pd.read_excel("C:/Users/User/Desktop/UTP/UG/2nd Year/2nd Semester/Data Sceince/Project/test.csv.xlsx")


TARGET = 'NObeyesdad_enc'

X_train = train.drop(columns=[TARGET])
y_train = train[TARGET]
X_test  = test.drop(columns=[TARGET])
y_test  = test[TARGET]

# ── Train Model ────────────────────────────────────────────────────────
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# ── Evaluate ───────────────────────────────────────────────────────────
predictions = model.predict(X_test)
accuracy    = accuracy_score(y_test, predictions)

print("=" * 50)
print("        RANDOM FOREST — Results")
print("=" * 50)
print(f"  Overall Accuracy : {accuracy:.2%}")
print(f"  n_estimators     : 100")
print(f"  Training samples : {len(train)}")
print(f"  Test samples     : {len(test)}")
print("=" * 50)
print("\nClassification Report:\n")
print(classification_report(y_test, predictions))
```

(***Figure 3.3:*** *Implementation of Random Forest Model*)

The code loads the pre-processed train and test datasets, trains a Random Forest ensemble of 100 individual decision trees, and evaluates its collective predictions against the full holdout test set.

How the Model Works and Key Settings

A Random Forest is best understood as a committee of decision trees. Rather than relying on a single decision tree, which can be sensitive to minor changes in the data, it builds 100 separate trees, each trained on a slightly different random portion of the data. When predicting someone's obesity level, every tree casts its own vote, and the category with the most votes wins. This democratic approach makes the model far more stable and accurate

than any individual tree could be on its own. The following settings were configured for this project:

- **n_estimators=100:** This defines how many individual decision trees are built and voted across. Using 100 trees ensures the ensemble captures a wide variety of patterns from the data, leading to a more reliable and consistent final prediction.
- **random_state=42:** This sets a fixed starting point so that the random sampling used to build each tree is reproducible, guaranteeing the same results every time the code is run.

Evaluation Strategy

The model's performance was evaluated by testing it against the unseen holdout dataset (418 samples). The predictions were then compared to the actual target values to generate the final accuracy, precision, recall, and F1-scores.

- **Overall Test Accuracy: 87.95%**

```
==================================================
        RANDOM FOREST  Results
==================================================
  Overall Accuracy : 87.95%
  n_estimators     : 100
  Training samples : 1672
  Test samples     : 415
==================================================

Classification Report:

              precision    recall  f1-score   support

           0       0.91      0.91      0.91        54
           1       0.76      0.71      0.74        55
           2       0.89      0.83      0.86        58
           3       0.85      0.82      0.83        55
           4       0.88      0.89      0.89        75
           5       0.85      0.98      0.91        54
           6       0.98      1.00      0.99        64

    accuracy                           0.88       415
   macro avg       0.88      0.88      0.88       415
weighted avg       0.88      0.88      0.88       415
```

(***Figure 3.4:*** *Evaluation of Random Forest Model*)

6.2.3. Support Vector Machine (SVM)

Code Implementation

The model was built using the SVC (Support Vector Classifier) class from the scikit-learn library. The following code demonstrates the training and evaluation process:

```python
import pandas as pd
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, classification_report

# ── Load Data ──────────────────────────────────────────────────────────
train = pd.read_csv("train.csv")
test  = pd.read_csv("test.csv")

TARGET = 'NObeyesdad_enc'

# ── Full train.csv features & labels ──────────────────────────────────
X_train = train.drop(columns=[TARGET])
y_train = train[TARGET]

# ── Full test.csv features & labels ───────────────────────────────────
X_test = test.drop(columns=[TARGET])
y_test = test[TARGET]

# ── Train Model ────────────────────────────────────────────────────────
model = SVC(kernel='rbf', C=1.0, gamma='scale', random_state=42)
model.fit(X_train, y_train)

# ── Final Test on test.csv ─────────────────────────────────────────────
test_predictions = model.predict(X_test)
test_accuracy    = accuracy_score(y_test, test_predictions)

print("=" * 55)
print("           SVM — Final Test (test.csv)")
print("=" * 55)
print(f"  Overall Accuracy  : {test_accuracy:.2%}")
print(f"  kernel            : RBF")
print(f"  C                 : 1.0")
print(f"  gamma             : scale")
print(f"  Training samples  : {len(X_train)}")
print(f"  Test samples      : {len(X_test)}")
print("=" * 55)
print("\nClassification Report (test.csv):\n")
print(classification_report(y_test, test_predictions))
```

(***Figure 3.5:*** *Implementation of SVM Model*)

The code loads the pre-processed train and test datasets, trains an SVM classifier using an RBF kernel, and evaluates its predictions the test set dataset.

How the Model Works and Key Settings

An SVM works by drawing the most optimal boundary line, called a hyperplane, between different obesity categories in the data. The clever part is that it uses a technique called the **RBF (Radial Basis Function) kernel**, which allows it to draw these boundaries in curved, complex shapes rather than just straight lines. This is critical for this dataset, because the lifestyle habits separating, say, "Overweight Level 1" from "Overweight Level 2" are not neatly divided. The SVM searches specifically for the boundary that creates the widest possible gap (the *margin*) between categories, making it more confident and precise in its classifications. The following settings were configured for this project:

- **kernel='rbf':** This instructs the model to use the Radial Basis Function, enabling it to draw non-linear, curved decision boundaries. It was chosen because the seven obesity categories in this dataset are not linearly separable; a straight line alone cannot cleanly divide them.
- **C=1.0 (Regularization Parameter):** This setting controls the trade-off between fitting the training data perfectly and keeping the decision boundary smooth and generalizable. A value of 1.0 is the balanced default, avoiding both overfitting and underfitting.
- **gamma='scale':** This controls how far the influence of a single data point reaches when drawing boundaries. Setting it to 'scale' automatically adjusts this value based on the number of features in the dataset, making it an adaptive and reliable choice.
- **random_state=42:** Ensures the model produces fully reproducible results across every run.

Evaluation Strategy

The model's performance was evaluated by testing it against the unseen holdout dataset (418 samples). The predictions were then compared to the actual target values to generate the final accuracy, precision, recall, and F1-scores.

- **Overall Test Accuracy: 74.46%**

```
=======================================================
          SVM  Final Test (test.csv)
=======================================================
  Overall Accuracy  : 74.46%
  kernel            : RBF
  C                 : 1.0
  gamma             : scale
  Training samples  : 1672
  Test samples      : 415
=======================================================

Classification Report (test.csv):

              precision    recall  f1-score   support

           0       0.71      0.87      0.78        54
           1       0.64      0.42      0.51        55
           2       0.78      0.55      0.65        58
           3       0.65      0.60      0.62        55
           4       0.72      0.76      0.74        75
           5       0.68      0.98      0.80        54
           6       1.00      1.00      1.00        64

    accuracy                           0.74       415
   macro avg       0.74      0.74      0.73       415
weighted avg       0.75      0.74      0.73       415
```

(***Figure 3.6:*** *Evaluation of SVM Model*)

6.2.4. K-Nearest Neighbors

Code Implementation

The model was built using the KneighborsClassifier class from the scikit-learn library. The following code demonstrates the training and evaluation process:

```python
import pandas as pd
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, classification_report

# ── Load Data ──────────────────────────────────────────────────────────
# Edit the filepath to your personal need so that the algorithm can actually find it
train = pd.read_excel("C:/Users/User/Downloads/DS Project/Model-Data-Code/train.csv.xlsx")
test  = pd.read_excel("C:/Users/User/Downloads/DS Project/Model-Data-Code/test.csv.xlsx")

TARGET = 'NObeyesdad_enc'

X_train = train.drop(columns=[TARGET])
y_train = train[TARGET]
X_test  = test.drop(columns=[TARGET])
y_test  = test[TARGET]

# ── Train Model ────────────────────────────────────────────────────────
model = KNeighborsClassifier(n_neighbors=5)
model.fit(X_train, y_train)

# ── Evaluate ───────────────────────────────────────────────────────────
predictions = model.predict(X_test)
accuracy    = accuracy_score(y_test, predictions)

print("=" * 50)
print("            KNN — Results")
print("=" * 50)
print(f"  Overall Accuracy : {accuracy:.2%}")
print(f"  k (neighbours)   : 5")
print(f"  Training samples : {len(train)}")
print(f"  Test samples     : {len(test)}")
print("=" * 50)
print("\nClassification Report:\n")
print(classification_report(y_test, predictions))
```

(***Figure 3.7:*** *Implementation of KNN Model*)

The code loads the pre-processed train and test datasets, trains a KNN classifier set to evaluate the 5 nearest neighbors, and evaluates its predictions against the full holdout test set.

How the Model Works and Key Settings

KNN does not learn a formula like most other models. Instead, when it needs to classify a new person, it looks through the entire training dataset and finds the 5 individuals whose lifestyle habits are mathematically most similar to that person. It then simply votes whichever obesity category appears most among those 5 neighbors becomes the prediction. Think of it as asking, *"Who are the 5 people most similar to this individual, and what weight category do they fall into?"* The following setting was configured for this project:

- **n_neighbors=5:** This defines how many "neighbor" individuals the model looks at before making a decision. A value of 5 was chosen as it strikes a practical balance; small enough to remain sensitive to local patterns in the data, but large enough to avoid being misled by a single unusual data point.

Evaluation Strategy

The model's performance was evaluated by testing it against the unseen holdout dataset (418 samples). The predictions were then compared to the actual target values to generate the final accuracy, precision, recall, and F1-scores.

- **Overall Test Accuracy: 73.98%**

```
==================================================
          KNN  Results
==================================================
  Overall Accuracy : 73.98%
  k (neighbours)   : 5
  Training samples : 1672
  Test samples     : 415
==================================================

Classification Report:

              precision    recall  f1-score   support

           0       0.74      0.93      0.82        54
           1       0.52      0.20      0.29        55
           2       0.63      0.64      0.63        58
           3       0.71      0.62      0.66        55
           4       0.72      0.79      0.75        75
           5       0.78      0.96      0.86        54
           6       0.91      1.00      0.96        64

    accuracy                           0.74       415
   macro avg       0.71      0.73      0.71       415
weighted avg       0.72      0.74      0.72       415
```

(***Figure 3.8:*** *Evaluation of KNN Model*)

6.2.5. Logistics Regression

Code Implementation

The model was built using the LogisticRegression class from the scikit-learn library. The following code demonstrates the training and evaluation process:

```python
import pandas as pd
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report

# ── Load Data ──────────────────────────────────────────────────────────
# Edit the filepath to your personal need so that the algorithm can actually find it
train = pd.read_excel("C:/Users/User/Downloads/DS Project/Model-Data-Code/train.csv.xlsx")
test  = pd.read_excel("C:/Users/User/Downloads/DS Project/Model-Data-Code/test.csv.xlsx")

TARGET = 'NObeyesdad_enc'

# ── Full train.csv features & labels ──────────────────────────────────
X_train = train.drop(columns=[TARGET])
y_train = train[TARGET]

# ── Full test.csv features & labels ───────────────────────────────────
X_test = test.drop(columns=[TARGET])
y_test = test[TARGET]

# ── Train Model ────────────────────────────────────────────────────────
model = LogisticRegression(max_iter=1000, random_state=42, solver='lbfgs')
model.fit(X_train, y_train)

# ── Final Test on test.csv ─────────────────────────────────────────────
test_predictions = model.predict(X_test)
test_accuracy    = accuracy_score(y_test, test_predictions)

print("=" * 55)
print("     LOGISTIC REGRESSION — Final Test (test.csv)")
print("=" * 55)
print(f"  Overall Accuracy  : {test_accuracy:.2%}")
print(f"  solver            : lbfgs")
print(f"  max_iter          : 1000")
print(f"  Training samples  : {len(X_train)}")
print(f"  Test samples      : {len(X_test)}")
print("=" * 55)
print("\nClassification Report (test.csv):\n")
print(classification_report(y_test, test_predictions))
```

(***Figure 3.9:*** *Implementation of Logistics Regression Model*)

The code loads the pre-processed train and test datasets, trains a Logistic Regression classifier using the lbfgs solver, and evaluates its predictions against the test data set.

How the Model Works and Key Settings

Logistic Regression works like a weighing scale. It takes a person's lifestyle features, such as how often they eat vegetables or how physically active they are, and assigns a mathematical weight to each one. It then combines all these weights to calculate the probability that the person belongs to each of the seven obesity categories, selecting whichever category scores the highest probability as its final answer. The following settings were configured for this project:

- **solver='lbfgs':** This is the optimization algorithm that guides how the model learns its weights during training. The lbfgs solver was selected because it is efficient and natively handles multiclass classification problems, making it a well-suited choice for predicting across all seven obesity levels simultaneously.
- **max_iter=1000:** This controls the maximum number of learning cycles the model is allowed to run before stopping. It was raised from the default to 1,000 to give the model enough time to fully converge and learn from the training data without prematurely stopping.
- **random_state=42:** This sets a fixed starting point to ensure the model produces consistent, reproducible results every time the code is run.

Evaluation Strategy

The model's performance was evaluated by testing it against the unseen holdout dataset (418 samples). The predictions were then compared to the actual target values to generate the final accuracy, precision, recall, and F1-scores.

- **Overall Test Accuracy: 58.07%**

```
=======================================================
     LOGISTIC REGRESSION — Final Test (test.csv)
=======================================================
  Overall Accuracy  : 58.07%
  solver            : lbfgs
  max_iter          : 1000
  Training samples  : 1672
  Test samples      : 415
=======================================================

Classification Report (test.csv):

              precision    recall  f1-score   support

           0       0.58      0.61      0.59        54
           1       0.43      0.29      0.35        55
           2       0.53      0.34      0.42        58
           3       0.52      0.27      0.36        55
           4       0.53      0.61      0.57        75
           5       0.48      0.87      0.62        54
           6       0.91      1.00      0.96        64

    accuracy                           0.58       415
   macro avg       0.57      0.57      0.55       415
weighted avg       0.57      0.58      0.56       415
```

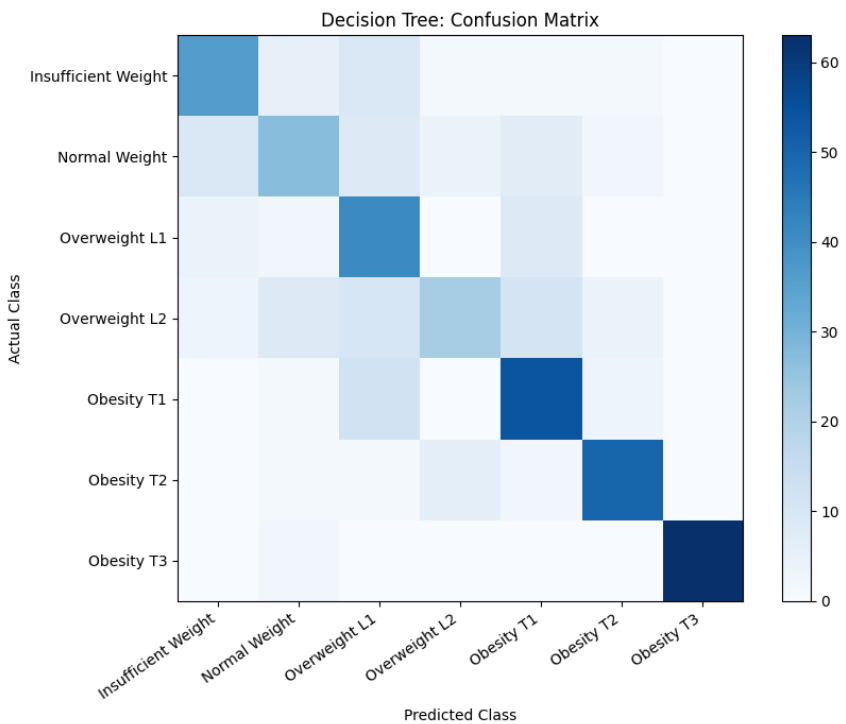
(***Figure 3.10:*** *Evaluation of Logistics Regression Model*)

7. Results, Findings and Insights

7.1. Decision Tree

Overall Performance

The Decision Tree model achieved a final accuracy of **69.64%**. While it does a decent job of finding general patterns in the lifestyle data, the visualizations show that it has clear strengths and weaknesses depending on which weight category it is trying to predict.



***(Figure 4.1:** Confusion matrix for the Decision Tree model)*

Where the Model Succeeds: Severe Obesity

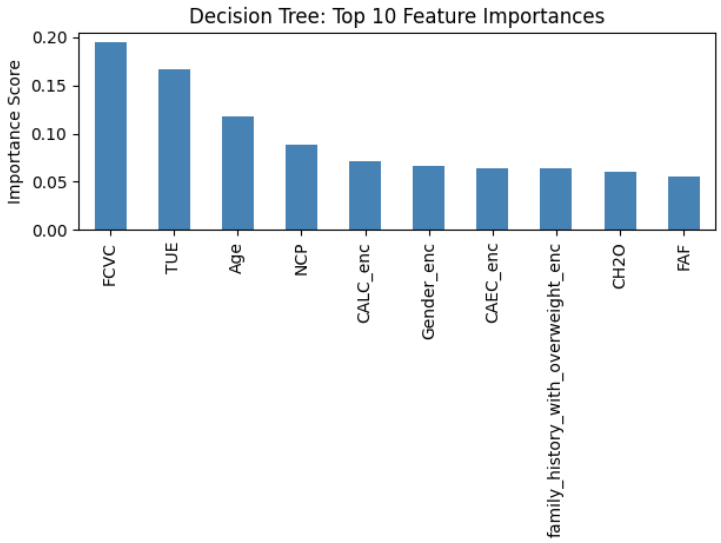
Looking at the **Confusion Matrix**, the darkest blue square on the entire board is in the bottom right corner, which represents **Obesity Type 3**. There are almost no misclassified squares in its row or column.

- **The Insight:** The model is nearly perfect at identifying people in the most severe obesity category. This tells us that individuals in this group share a very distinct, easily recognizable lifestyle profile that stands out clearly from the rest of the dataset.

Where the Model Struggles: The "Messy Middle"

In the center of the Confusion Matrix (covering Normal Weight, Overweight L1, and Overweight L2), the squares are much paler, and the blue color is scattered across multiple boxes.

- **The Insight:** The model frequently mixes these people up. For example, it often guesses someone is "Overweight L1" when they are actually "Normal Weight." This proves that the daily habits (like diet and exercise) of people in these middle categories are very similar. The boundaries between being a normal weight and slightly overweight are too blurry for this simple model to separate perfectly.



(Figure 4.2:** Barchart of most important features in Decision Tree model**)

Feature Importance:

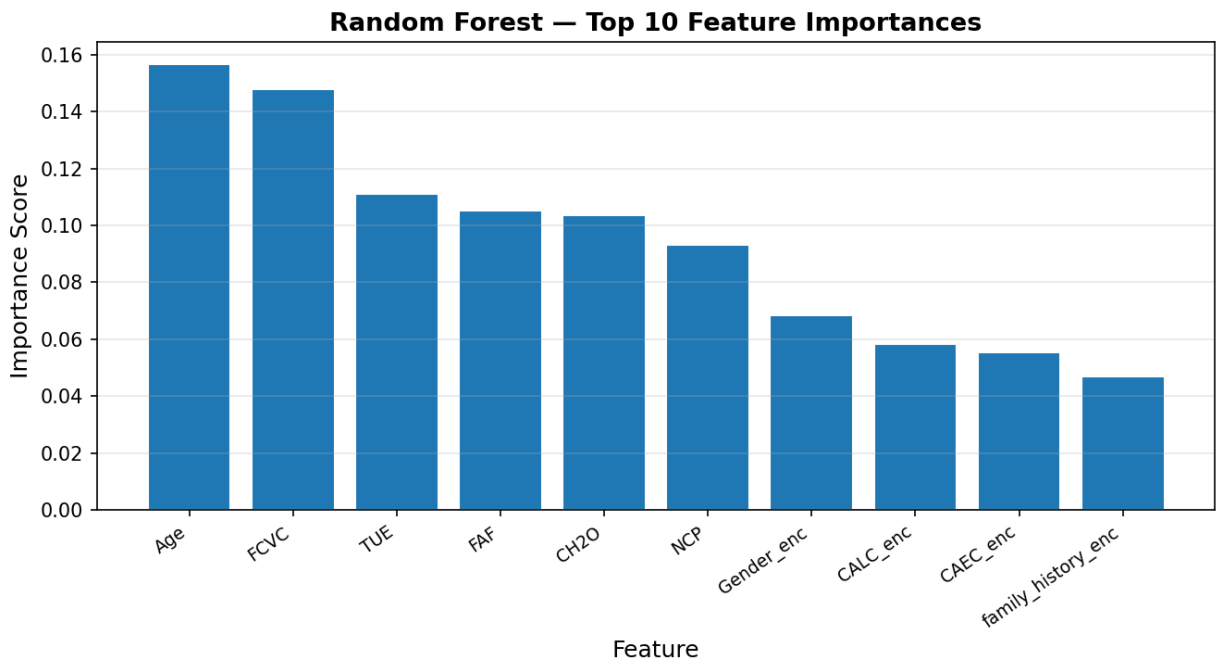
The **Feature Importance** bar chart shows us exactly which lifestyle habits the model relied on most to make its predictions. The taller the bar, the more important the habit:

- **Diet and Screen Time Matter Most:** The two tallest bars by far are **FCVC** (how often a person eats vegetables) and **TUE** (how much time they spend looking at technology/screens). These two habits are the heaviest drivers of weight classification.
- **Genetics Matter Less:** Interestingly, the bar for family_history_with_overweight is much shorter, ranking near the bottom of the top 10.
- **The Insight:** According to this specific model, your immediate, daily choices, specifically eating vegetables and limiting sedentary screen time, are much stronger predictors of your weight class than your family genetics.

7.2. Random Forest

Overall Performance

The Random Forest model achieved a final test accuracy of **87.95%**, making it the strongest-performing model in this project. It builds 100 independent decision trees, each trained on a random portion of the data, and combines their votes into a single, robust prediction; an ensemble approach that effectively smooths out the errors of any individual tree.



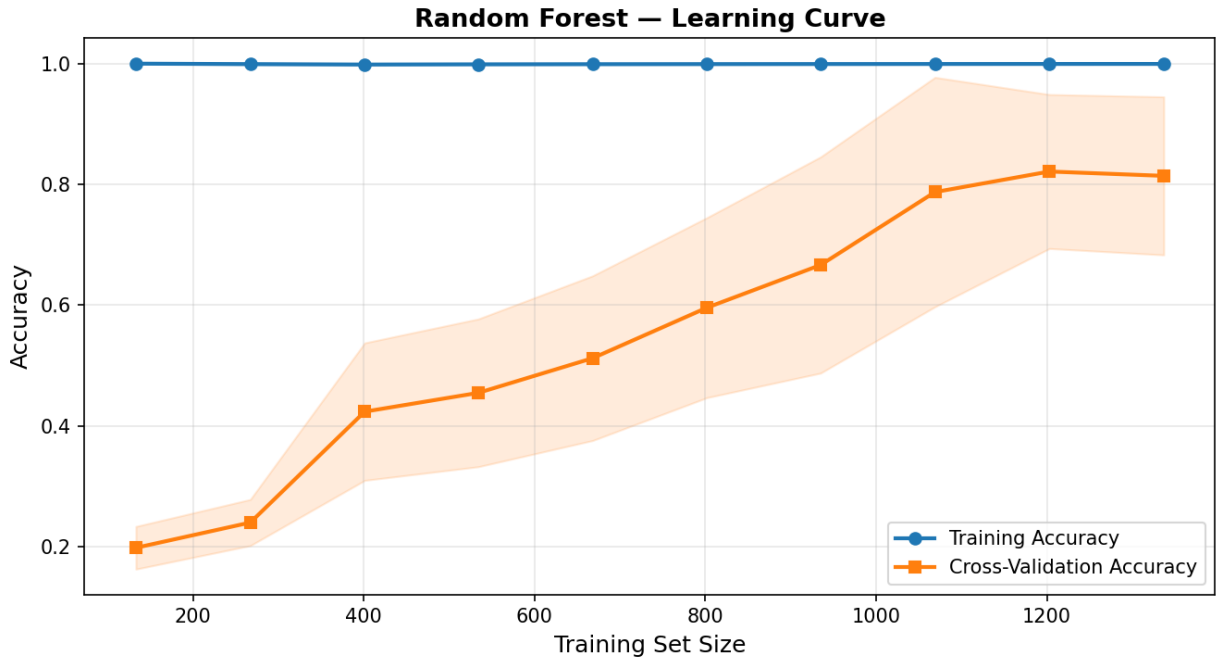
(Figure 4.3:** Barchart of most important features in Random Forest model**)

Where the Model Succeeds: Learning What Matters Most

The **Feature Importance Chart** ranks each lifestyle feature by how much it contributed to the model's classification decisions across all 100 trees. The taller the bar, the more the model relied on that feature.

- **The Insight:** The chart shows that dietary habits like **FCVC (vegetable consumption frequency)** and **NCP (number of main meals)**, alongside **TUE (technology/screen time)** and **Age** (a biological control), are the most dominant predictors. Notably, transportation mode features rank at the bottom, suggesting they add minimal predictive value for obesity classification in this dataset. The Feature Importance chart is the definitive graph for showcasing Random Forest's strength because it is a unique output that only tree-based ensemble models can produce — it offers direct, quantified insight

into which real-world behaviors are mathematically driving obesity predictions, bridging the gap between model performance and actionable health insight.



***(Figure 4.4:** Learning curve for the Random Forest model)*

Where the Model Struggles: A Ceiling Despite More Data

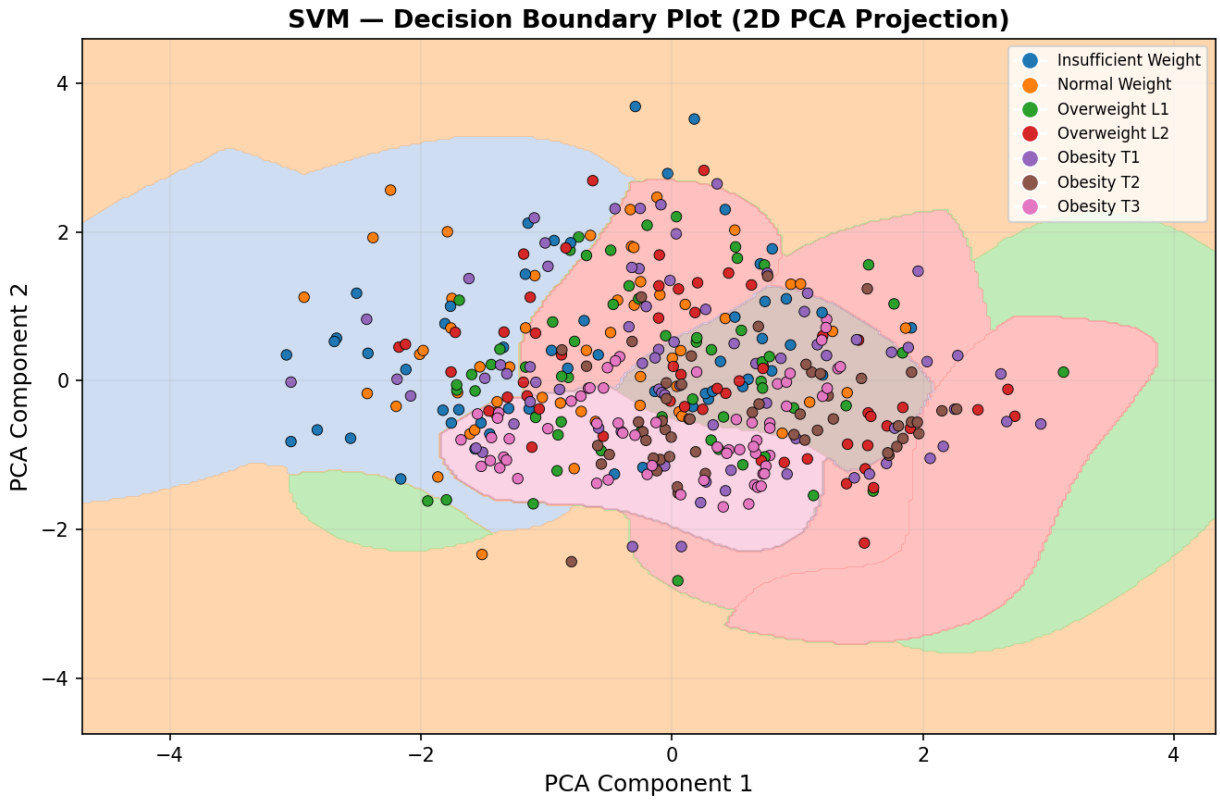
The **Learning Curve** plots how the model's training accuracy and cross-validation accuracy evolve as the size of the training dataset increases from 10% to 100%.

- **The Insight:** The training accuracy line stays near-perfect throughout, while the cross-validation accuracy climbs early but plateaus and stabilizes well below it. The persistent gap between the two lines is the key signal — it indicates that the model has a mild overfitting tendency, meaning it learns the training data slightly too well and does not generalize as cleanly to unseen data. Furthermore, the flattening of the validation curve toward the right shows that simply adding more training data yields diminishing returns. This reveals that the model has reached its practical performance ceiling under its current configuration, and further improvements would require deeper hyperparameter tuning rather than more data. The Learning Curve is the ideal graph to expose this because it makes the overfitting gap visible and measurable.

7.3. Support Vector Machine (SVM)

Overall Performance

The SVM model achieved a final test accuracy of **74.46%**. Using an RBF kernel, it searches for the optimal curved boundary, called a hyperplane, that maximizes the margin of separation between each obesity class, making it a mathematically precise model that excels when classes are well-defined but struggles when lifestyle profiles of adjacent categories overlap closely.



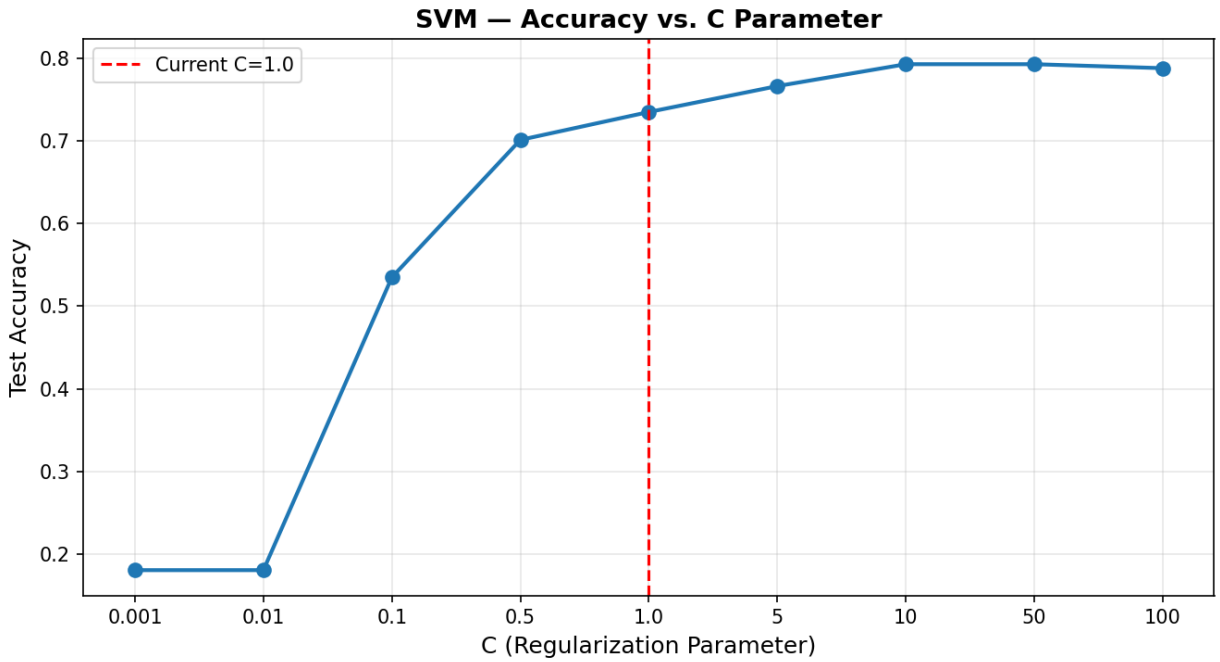
*(**Figure 4.5:** 2D PCA decision boundary plot for the SVM model)*

Where the Model Succeeds: Sharp, Confident Boundaries

The **Decision Boundary Plot** projects the data into 2 dimensions using PCA and shades the classification regions the SVM model has carved out for each obesity class.

- **The Insight:** The plot shows smooth, continuously shaped boundaries, which is a direct visual result of the RBF kernel's ability to draw non-linear, curved decision surfaces. The shaded regions for extreme classes such as Obesity Type 3 are compact and precise, with very few misclassified points falling outside their designated region. This directly demonstrates the SVM's core mathematical strength: it does not just find any boundary between classes; it finds the one that maximizes the gap between them. The Decision

Boundary Plot is the most fitting visualization for SVM precisely because SVM is fundamentally a boundary-drawing model, and this graph makes that geometric strength directly visible and intuitive.



*(**Figure 4.6:** SVM model test accuracy across different values of the regularization C parameter)*

Where the Model Struggles: Sensitive to Regularization

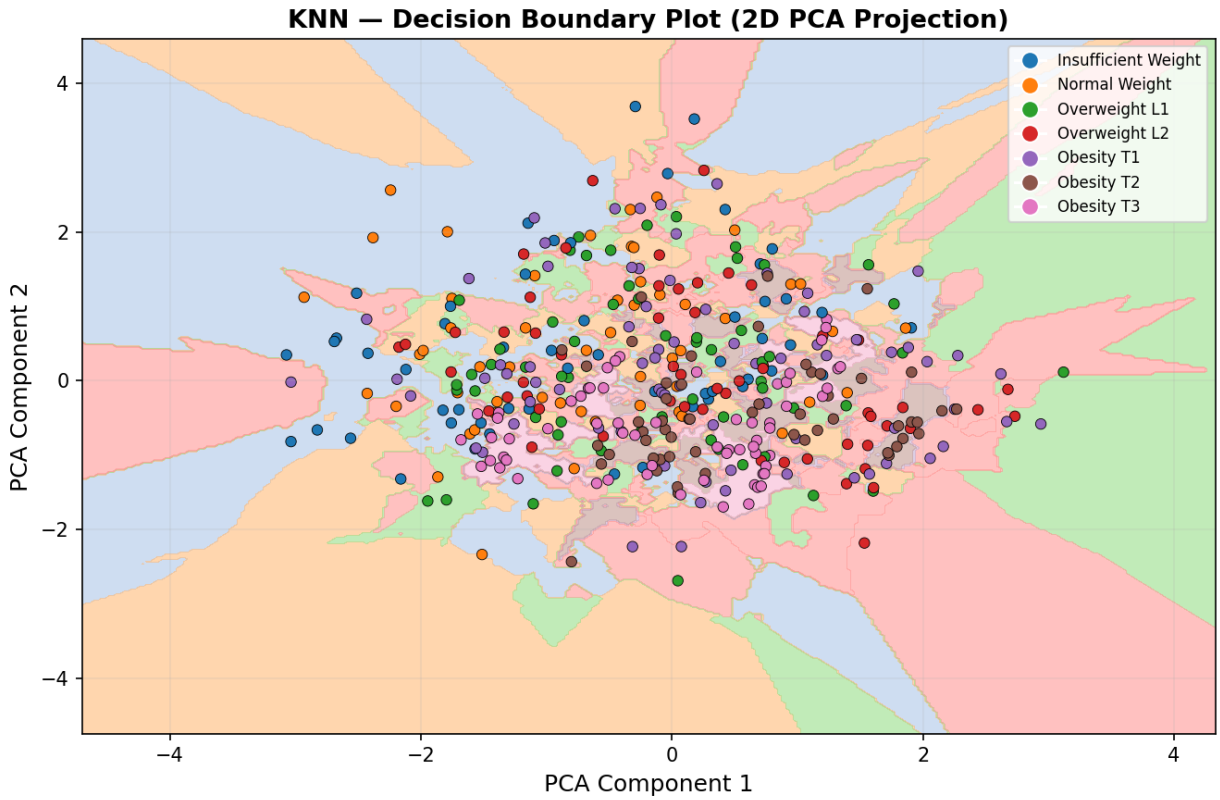
The **Accuracy vs. C Parameter Plot** shows how the model's test accuracy shifts as the regularization value C is adjusted across a wide range, from very small (0.001) to very large (100).

- **The Insight:** The plot reveals a pronounced accuracy gap between low and high C values, with performance being poor at very small C values and stabilizing only within a specific mid-range window. The current setting of C=1.0, while reasonable, does not fall at the point of peak accuracy on the curve. This demonstrates a well-known practical weakness of SVM — its performance is not robust across different C settings and requires careful, deliberate tuning to reach its true potential. Without this tuning, the model risks either drawing a boundary that is too rigid (high C, overfitting) or too loose (low C, underfitting). The Accuracy vs. C plot is the most direct and transparent way to expose this sensitivity, translating an abstract mathematical parameter into a concrete, visible consequence on predictive accuracy.

7.4. K-Nearest Neighbors (KNN)

Overall Performance

The KNN model achieved a final test accuracy of **73.98%**. It classifies each individual's obesity level by identifying the 5 most mathematically similar people in the training data and voting on the majority class. KNN is an intuitive, neighborhood-based approach that performs reasonably well on clearly clustered classes but struggles where lifestyle profiles overlap.



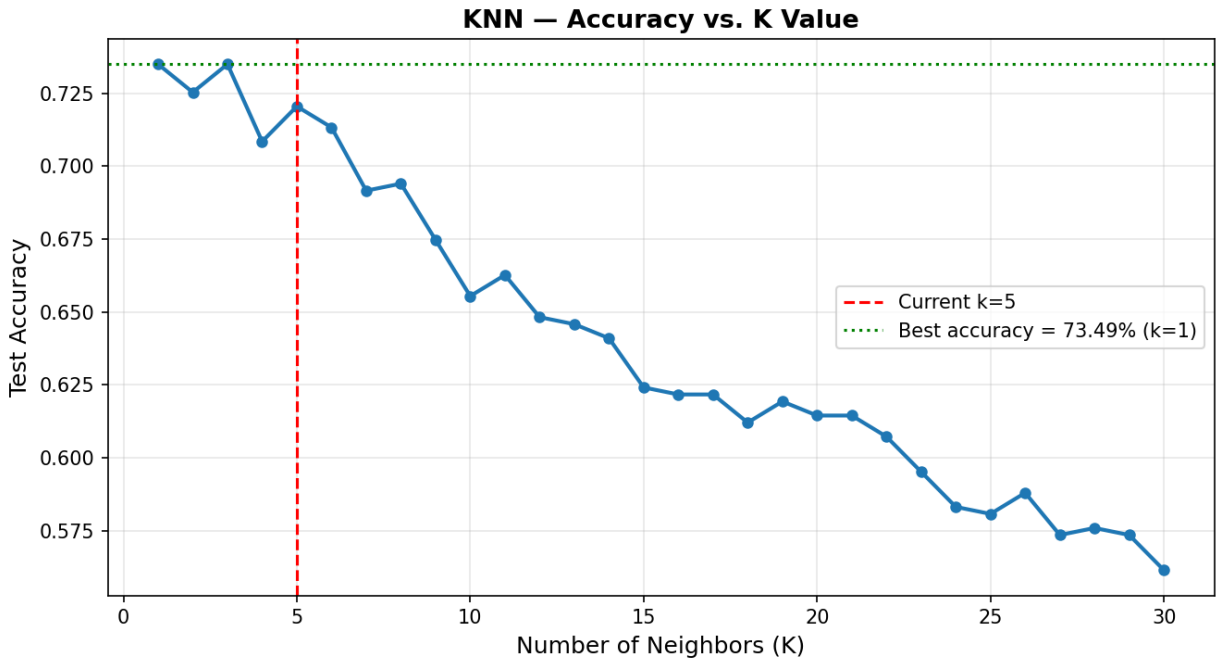
(***Figure 4.7:*** *2D PCA decision boundary plot for the KNN model)*

Where the Model Succeeds: Clean Local Clusters

The **Decision Boundary Plot** reduces the full dataset down to 2 dimensions using PCA and overlays the regions that the KNN model has assigned to each obesity class. Each shaded region represents the area where the model will predict a given class.

- **The Insight:** The plot reveals distinct, well-separated shaded regions on the left and right extremes of the PCA space, corresponding to the Insufficient Weight and Obesity Type 3 classes. The data points within these regions are dense and tightly clustered, meaning the individuals in these categories share very similar lifestyle profiles. This directly demonstrates KNN's core strength. When a natural cluster exists in the data, the model

reliably wraps a clean boundary around it. The Decision Boundary Plot complements KNN specifically because KNN is inherently a spatial, distance-based model, and this graph visualizes exactly how it partitions space.



*(**Figure 4.8:** Test accuracy of the KNN model across different values of K)*

Where the Model Struggles: Fragile to K Selection

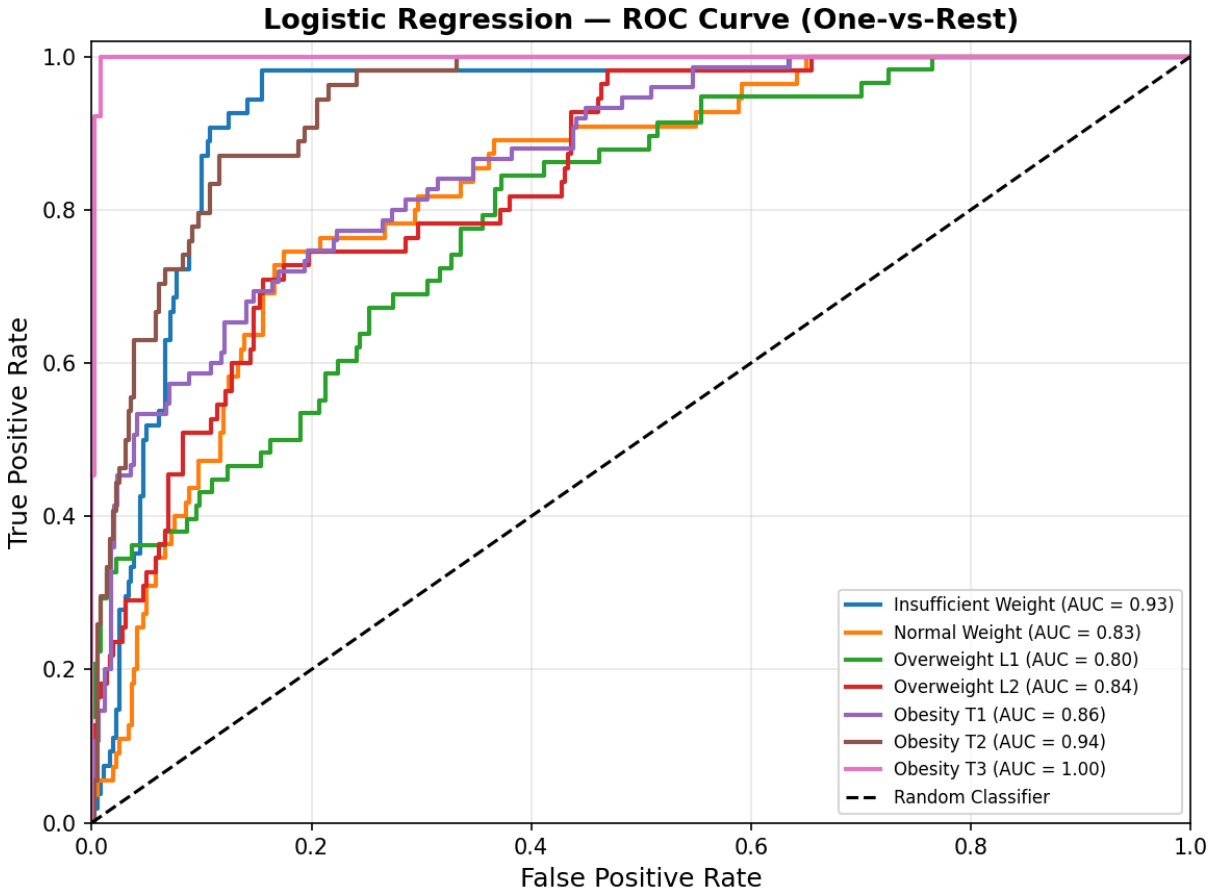
The **Accuracy vs. K Value Plot** shows how the model's test accuracy changes as the number of neighbors evaluated (K) increases from 1 to 30.

- **The Insight:** The plot shows significant fluctuation in accuracy across different K values, with performance peaking at a specific K and then declining as K grows too large. The current setting of K=5 does not land on the highest achievable accuracy, showing that the model's performance is fragile and highly sensitive to this single hyperparameter. This is a critical practical weakness. Without systematic tuning, the model is likely not operating at its full potential. The Accuracy vs. K plot is the most direct way to expose this, as it translates an abstract hyperparameter into a concrete, visible impact on real-world prediction performance.

7.5. Logistics Regression

Overall Performance

The Logistic Regression model achieved a final test accuracy of **58.07%**. As the statistical baseline of this project, it attempts to separate all seven obesity classes using linear mathematical boundaries drawn from each individual's lifestyle features. This is an approach that proves sufficient only for the most extreme, well-separated categories in the dataset.



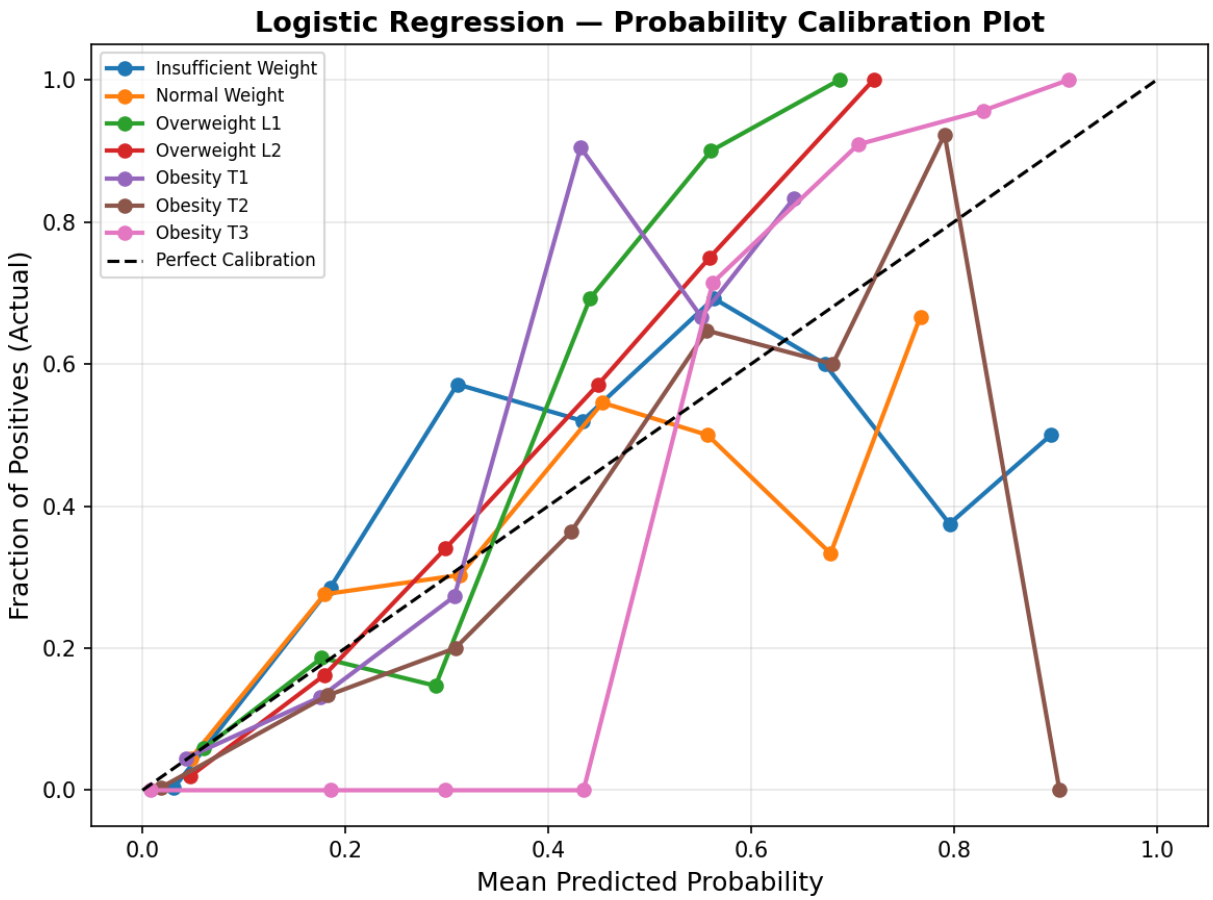
*(**Figure 4.9:** Receiver Operating Characteristic (ROC) curve for the Logistic Regression model)*

Where the Model Succeeds: Separating the Extremes

The **ROC Curve** plots the True Positive Rate against the False Positive Rate for each obesity class individually. A curve that climbs steeply toward the top-left corner indicates a model that separates that class well, regardless of overall accuracy.

- **The Insight:** Despite a modest overall accuracy, the ROC curves for **Obesity Type 3** (AUC ≈ 1.00) and **Insufficient Weight** sit far above the diagonal, demonstrating that the

model reliably distinguishes individuals in the most extreme weight categories. This tells us that the lifestyle habits of these two groups are so distinct and linearly separable that even a simple model like Logistic Regression can draw a clean boundary around them. The ROC Curve complements Logistic Regression specifically because it evaluates class-level separability independently of the threshold, which is the exact strength of a probability-based model like this one.



(***Figure 4.10***: *Probability calibration plot for the Logistic Regression model*)

Where the Model Struggles: Overconfident in the Middle

The **Probability Calibration Plot** compares what the model *predicted* the probability would be against what actually happened. A perfectly calibrated model would follow the straight diagonal line exactly.

- **The Insight:** The curves for the middle categories, which are Normal Weight, Overweight L1, and Overweight L2, deviate noticeably from the perfect calibration diagonal. In several ranges, the model predicts a high probability of belonging to these classes, but the actual outcome does not follow. This reveals that Logistic Regression is not just inaccurate in the middle. It is also overconfident and poorly calibrated in those predictions. The Calibration Plot is the ideal graph to expose this weakness because it directly interrogates the quality of the model's probability estimates, which is the core output mechanism of Logistic Regression.

8. Limitations & Future Improvements

8.1. Limitations

i. Small Dataset from a Single Source.

The dataset contained only 2,111 individuals from one Kaggle source, specifically people from Mexico, Peru, and Colombia, and after cleaning, only around 1,672 were used for training. While the dataset covers 16 lifestyle and health-related features including eating habits, family history, and physical condition across seven obesity classifications, it remains geographically limited to just three Latin American countries. This means the behavioral patterns and lifestyle habits captured here may not accurately represent people from different parts of the world, such as Asia, Europe, or Africa, where diets, cultural habits, and physical activity norms can be very different. Essentially, what works for this dataset may not automatically work everywhere else.

ii. Data Captured at One Point in Time.

All the information in this dataset was collected once, like a photograph. But, in reality, obesity develops slowly over time through years of habits. Because we only have a single snapshot of each person's lifestyle, the models cannot detect how someone's habits might be changing. For example, someone who recently started exercising more or eating worse. This limits how realistic the predictions are in a real-world setting.

iii. Hard to Tell Apart the "In-Between" Weight Categories.

All five models consistently struggled to correctly classify people who fell in the middle weight categories, which are Normal Weight, Overweight Level I, and Overweight Level II. This is because people in these groups tend to have very similar daily habits, making it genuinely difficult, even for a machine learning model, to tell them apart. The features currently available in the dataset simply do not capture enough detail to clearly separate these groups.

iv. Imbalanced Independent Features.

During the exploratory data analysis phase, three lifestyle features were found to be severely imbalanced. This means that one answer made up the overwhelming majority of responses. Specifically, over 97% of individuals did not smoke (SMOKE), over 95% did not count calories (SCC), and over 97% used passive transportation. Because almost everyone gave the same answer for these three features, they provide very little useful information to distinguish between different obesity levels. While this does not directly harm the models since these are input features rather than the target variable, it does mean these three features are contributing very little predictive value and may be taking up space that could be better used by more informative variables.

v. Height and Weight Were Removed on Purpose.

To make sure the models learned from lifestyle habits rather than just calculating body size, Height and Weight were deliberately left out. While this was the right decision for the goals of this study, it does mean the models are missing two pieces of information that doctors would normally have access to in a real clinical setting. This intentional removal places a ceiling on how accurate the models can realistically become.

8.2. Future Improvements

i. Add More Useful Features

Since the current features struggle to separate the middle weight categories, future studies should consider collecting additional lifestyle information such as sleep quality, stress levels, meal timing or income level. These extra details could give the models the finer information they need to make better predictions for people who are borderline between weight categories.

ii. Address the Imbalanced Features

For the three severely imbalanced features, which are SMOKE, SCC, and Passive Transportation, future work could either collect more diverse data that includes a wider range of responses or consider removing these features entirely if they continue to show little predictive value. Either approach would help ensure that every feature in the model is genuinely earning its place.

iii. Track People Over Time

Instead of collecting data once, future studies should follow the same individuals over months or years. This would allow the models to detect gradual changes in habits such as someone slowly becoming less active which is far more useful for predicting obesity risk early before it becomes a serious health problem.

iv. Use More Diverse Data

To make the findings applicable to a wider range of people, future datasets should include individuals from different countries, cultures, and age groups. A model trained on more diverse data is much more likely to work reliably for people from all walks of life, not just those represented in this particular dataset.

9. Conclusion

This project set out to answer a deceptively simple question; "Can a person's daily lifestyle habits alone predict their obesity risk?" Through a full data science lifecycle, spanning exploratory data analysis, rigorous data preparation and the training and evaluation of five distinct machine learning models, the answer is a meaningful and qualified yes.

The analytical journey began with a structured Kaggle dataset of 2,111 individuals, each described by 17 attributes capturing their dietary habits, physical activity levels, transportation choices and demographic information. After removing 24 duplicate records, engineering key features, applying Winsorization to remove extreme outliers and deliberately dropping Height and Weight to prevent target leakage, the dataset was distilled into 14 lifestyle features, ensuring that every model was forced to learn from how people *live*, not simply from how heavy they are.

Five classification algorithms were then trained and evaluated against an unseen holdout test set of 415 individuals. The results paint a clear and consistent picture:

Model	Test Accuracy
Logistic Regression	58.07%
Decision Tree	69.64%
KNN	73.98%
SVM	74.46%
Random Forest	**87.95%**

(***Table 5.1:*** *Test Accuracy for each training model*)

The **Random Forest** emerged as the standout model of this study, achieving a test accuracy of 87.95% which is nearly 14 percentage points ahead of its nearest competitors. Its method of aggregating the votes of 100 independent decision trees, proved to be the most effective strategy for navigating the complex, non-linear relationships that exist between lifestyle behaviors and obesity classification. Its Feature Importance analysis further revealed that Age, vegetable consumption frequency (FCVC), and physical activity frequency (FAF) were the most influential predictors.

A finding consistent across all five models was the sharp contrast in classification difficulty between the extreme and middle weight categories. Every model, regardless of its underlying algorithm, achieved near-perfect or perfect identification of Obesity Type III individuals, while consistently struggling to differentiate between Normal Weight, Overweight Level I and Overweight Level II. This tells us something important that goes beyond the models themselves: the lifestyle habits that separate someone who is slightly overweight from someone of normal weight are genuinely subtle, overlapping, and difficult to differentiate. Obesity, at its early stages, does not announce itself loudly in the data.

It is the sincere hope of this study that the insights generated here extend beyond the boundaries of an academic exercise. In a world where obesity rates continue to climb, carrying with them an escalating burden of chronic disease, healthcare costs and reduced quality of life. Tools that can identify at-risk individuals early, based purely on their daily behaviors, carry real value. The aspiration is that frameworks like this one may one day inform mobile health applications, clinical screening tools or public health policy engines that gently guide individuals toward awareness before a health crisis emerges. Every percentage point of predictive accuracy, in that context, represents a real person whose risk could be identified earlier.

Looking ahead, this study opens several meaningful opportunities for future improvements. The persistent difficulty in classifying middle-range weight categories across all five models suggests that the current dataset while valuable, may benefit from the incorporation of richer behavioral data, such as sleep quality, psychological stress indicators or longitudinal tracking of habit changes over time. There is also a compelling case for exploring deep learning architectures on larger, more diverse population datasets, particularly those that span different cultural dietary patterns and geographic regions to test whether the behavioral predictors identified here generalize beyond the demographics captured in this dataset. Finally, the integration of this predictive model into a real-time, interactive dashboard represents an impactful next step, to deploy and instrument for preventive healthcare.

10.References

1. Chirag Shah. (2020). *A hands-on introduction to data science* (1st ed.). Cambridge University Press.
2. GeeksforGeeks. (n.d.). *Supervised machine learning*. https://www.geeksforgeeks.org/machine-learning/supervised-machine-learning/
3. IBM. (n.d.). *Exploratory data analysis (EDA)*. https://www.ibm.com/think/topics/exploratory-data-analysis
4. IBM. (n.d.). *What is data cleaning?* https://www.ibm.com/think/topics/data-cleaning
5. World Health Organization. (2021). *Obesity and overweight*. https://www.who.int/news-room/fact-sheets/detail/obesity-and-overweight

11. Appendix

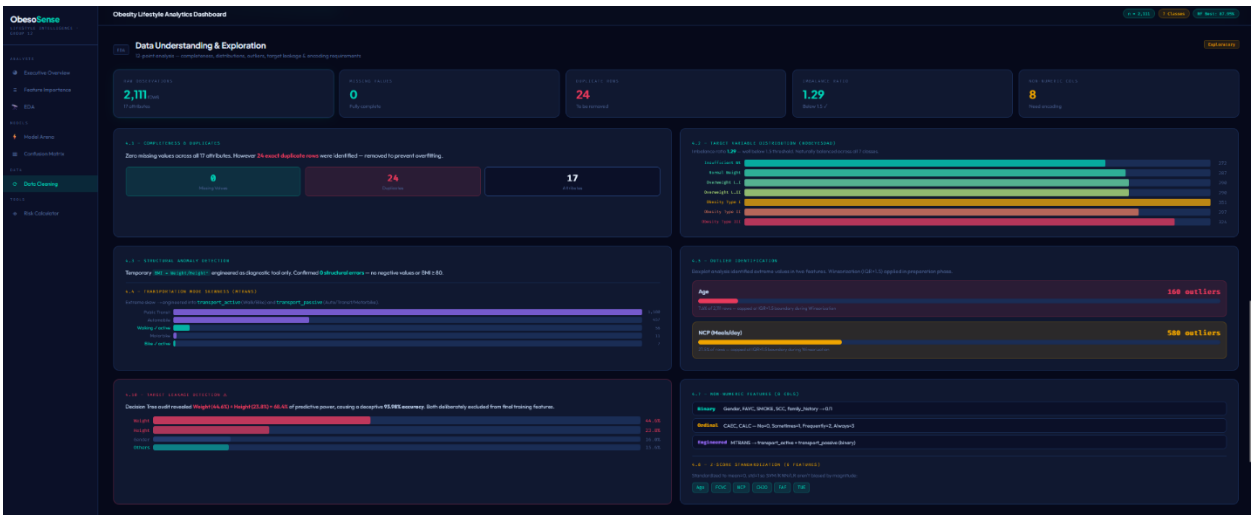
11.1 GitHub Repository

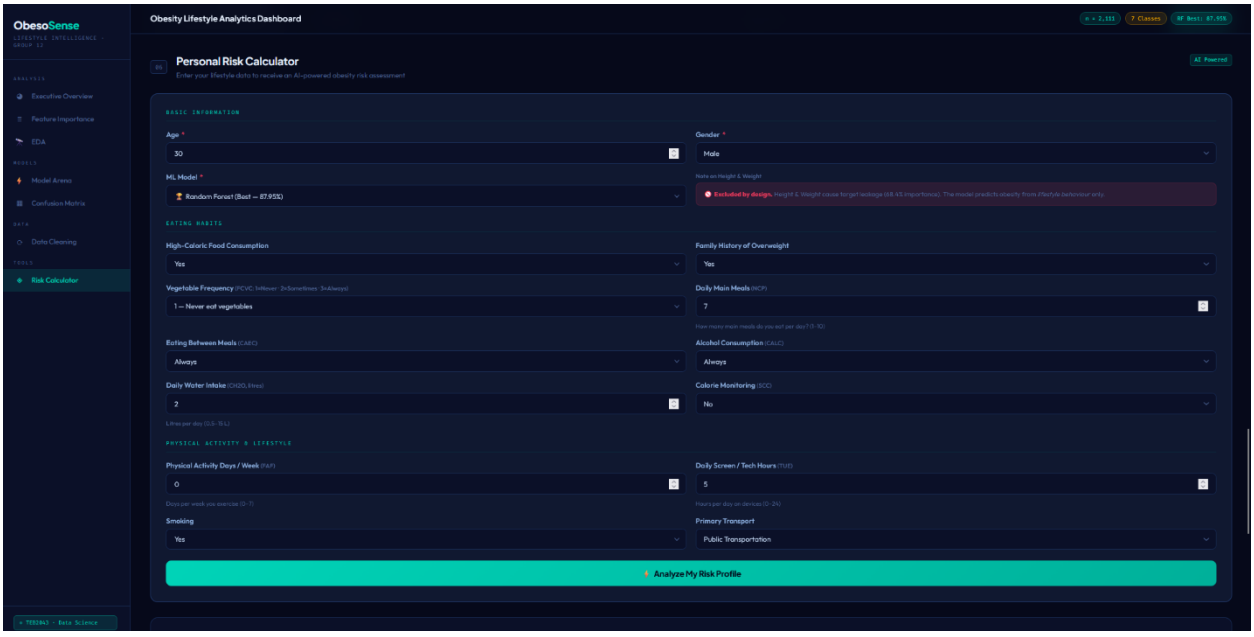
https://github.com/AmmirrulAmran/Data-Science-Project-on-Obesity-Prediction-Based-on-Lifestyle

(This Repository Contains code for Data Preparationa and Cleaning Machine Learning Models, Dashboard Code, Training and Test Dataset and Raw Dataset)

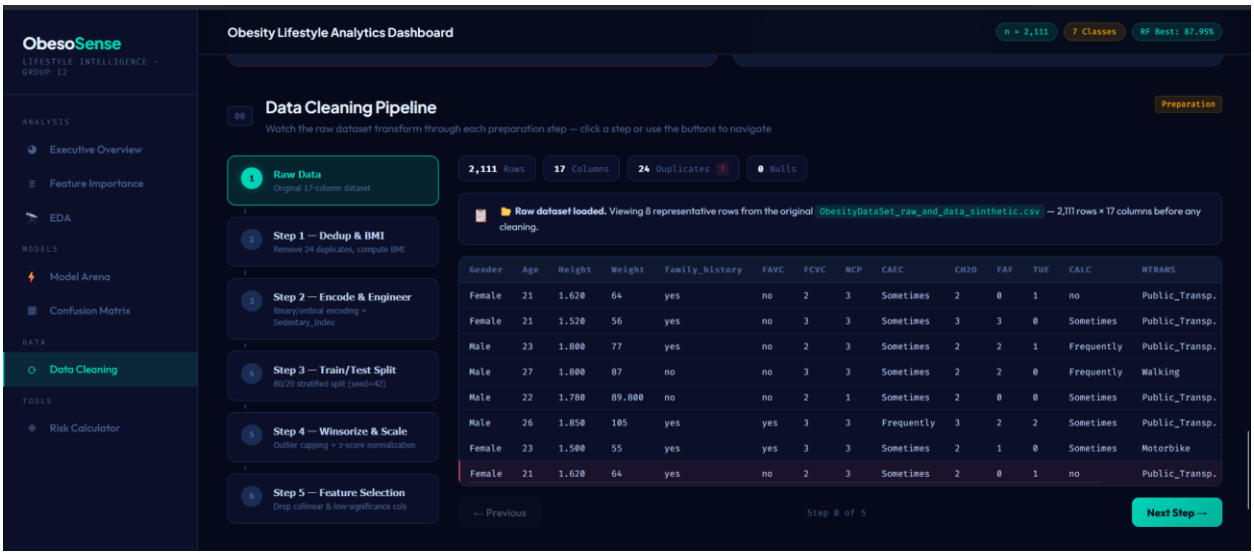
11.2 Dashboard Preview

ObesoSense
LIFESTYLE INTELLIGENCE · GROUP 12
Obesity Lifestyle Analytics Dashboard
n = 2,111
7 Classes
RF Best: 87.95%
ANALYSIS
Executive Overview
Feature Importance
EDA
MODELS
Model Arena
Confusion Matrix
DATA
Data Cleaning
TOOLS
Risk Calculator
TE82043 · Data Science
03
Feature Importance
Random Forest feature_importances_ — which habits drive obesity?
Diagnostic
01 Age 15.7%
02 Vegetable Freq. 15.1%
03 Tech Use Time ENGINEERED 10.5%
04 Water Intake 10.3%
05 Physical Activity 10.0%
06 Meal Frequency 9.0%
07 Gender 7.1%
08 Alcohol Freq. 6.2%
09 Snacking Freq. 5.5%
10 Family History 4.8%
11 High-Cal Food 3.0%
12 Calorie Monitor 1.4%
13 Passive Transport 0.7%
14 Smoking 0.6%
DECISION TREE VS RANDOM FOREST
Random Forest
Decision Tree
Age
etable Freq.
ech Use Time
Water Intake
cal Activity
al Frequency
Gender
0% 2% 4% 6% 8% 10% 12% 14% 16% 18%
04
Model Arena
Head-to-head performance: Test Accuracy vs Cross-Validation Accuracy
Predictive


ObesoSense
Obesity Lifestyle Analytics Dashboard
Data Understanding & Exploration
Exploratory
2,111
0
24
1.29
8
Zero missing values across all 17 attributes. However 24 exact duplicate rows were identified — removed to prevent overfitting.
Missing Values
Duplicates
17
Age
160 outliers
NCP (Meals/day)
588 outliers


ObesoSense
Obesity Lifestyle Analytics Dashboard
n = 2,111
7 Classes
RF Best: 87.95%
05
Personal Risk Calculator
Enter your lifestyle data to receive an AI-powered obesity risk assessment
AI Powered
BASIC INFORMATION
Age *
30
Gender *
Male
ML Model *
Random Forest (Best — 87.95%)
Note on Height & Weight
Excluded by design. Height & Weight cause target leakage (68.4% importance). The model predicts obesity from lifestyle behaviour only.
EATING HABITS
High-Caloric Food Consumption
Yes
Family History of Overweight
Yes
Vegetable Frequency (FCVC: 1=Never 2=Sometimes 3=Always)
1 — Never eat vegetables
Daily Main Meals (NCP)
3
Have many main meals do you eat per day? (1-10)
Eating Between Meals (CAEC)
Always
Alcohol Consumption (CALC)
Always
Daily Water Intake (CH2O, litres)
2
Litres per day (0.5–15 L)
Calorie Monitoring (SCC)
No
PHYSICAL ACTIVITY & LIFESTYLE
Physical Activity Days / Week (FAF)
0
Days per week you exercise (0–7)
Daily Screen / Tech Hours (TUE)
5
Hours per day on devices (0–24)
Smoking
Yes
Primary Transport
Public Transportation
Analyze My Risk Profile
TE82043 · Data Science

Executive Overview
Feature Importance
EDA
Model Arena
Confusion Matrix
Data Cleaning
Risk Calculator
Overweight Level II
The model identifies Overweight Level II with 21% confidence. Your sedentary index (5.00), snacking habit (Always), and screen time (5h/day) are primary drivers. Target 3–4 active days/week and reduce high-caloric food intake.
Low activity: only 0 day(s)/week exercise
High sedentary index: 5 (screen time vs activity)
Insufficient Weight 13.5%
Normal Weight 16.5%
Overweight Level I 20.0%
Overweight Level II 20.5%
Obesity Type I 17.5%
Obesity Type II 12.0%
Obesity Type III 0.0%


ObesoSense
LIFESTYLE INTELLIGENCE · GROUP 12
Obesity Lifestyle Analytics Dashboard
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ANALYSIS
Executive Overview
Feature Importance
EDA
MODELS
Model Arena
Confusion Matrix
DATA
Data Cleaning
TOOLS
Risk Calculator
Data Cleaning Pipeline
Preparation
Watch the raw dataset transform through each preparation step — click a step or use the buttons to navigate
Raw Data
Original 17-column dataset
Step 1 — Dedup & BMI
Remove 24 duplicates, compute BMI
Step 2 — Encode & Engineer
Binary/ordinal encoding + Sedentary_Index
Step 3 — Train/Test Split
80/20 stratified split (seed=42)
Step 4 — Winsorize & Scale
Outlier capping + z-score normalization
Step 5 — Feature Selection
Drop collinear & low-significance cols
2,111 Rows
17 Columns
24 Duplicates
0 Nulls
Raw dataset loaded. Viewing 8 representative rows from the original ObesityDataSet_raw_and_data_sinthetic.csv — 2,111 rows × 17 columns before any cleaning.
Gender Age Height Weight family_history FAVC FCVC NCP CAEC CH2O FAF TUE CALC MTRANS
Female 21 1.620 64 yes no 2 3 Sometimes 2 0 1 no Public_Transp.
Female 21 1.520 56 yes no 3 3 Sometimes 3 3 0 Sometimes Public_Transp.
Male 23 1.800 77 yes no 2 3 Sometimes 2 2 1 Frequently Public_Transp.
Male 27 1.800 87 no no 3 3 Sometimes 2 2 0 Frequently Walking
Male 22 1.780 89.800 no no 2 1 Sometimes 2 0 0 Sometimes Public_Transp.
Male 26 1.850 105 yes yes 3 3 Frequently 3 2 2 Sometimes Public_Transp.
Female 23 1.500 55 yes yes 3 3 Sometimes 2 1 0 Sometimes Motorbike
Female 21 1.620 64 yes no 2 3 Sometimes 2 0 1 no Public_Transp.
← Previous
Step 0 of 5
Next Step →